
Cosmology

Despite its name, the big bang theory is not really a theory of a bang at all. It is really only a theory of the aftermath of a bang.

—ALAN GUTH

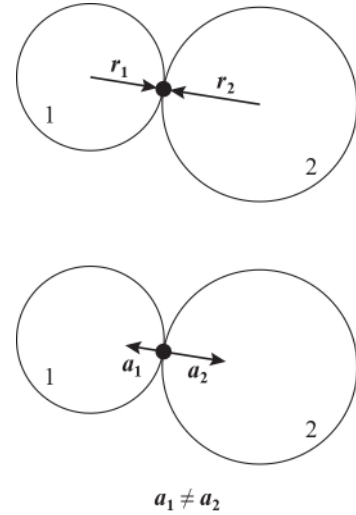
The subject of the origin of the Universe is irresistible to the scientist and layperson alike. How did the Universe originate? If a big bang, what went bang? What has happened along the way? Where are we headed? With its rigorous mathematical formulation of the large-scale geometry of spacetime, general relativity theory provides a complete conceptual apparatus for understanding the structure and evolution of the Universe. By the end of the 1920s, the observational astronomers had shown that we live in an expanding universe, anticipated theoretically in 1922 by the work of Alexander Friedmann. We are fortunate to live today in an era in which many scientifically precise answers to the great questions of cosmology are at hand. Moreover, while we need relativity to place ourselves on a properly rigorous footing, we can actually get quite far using very simple classical physics with hardly any relativity at all. This is an excellent way to begin our study. Let us start with some very Newtonian dynamics and see what there is to see. Then, knowing a bit of what to expect and where we might be headed, we will be in a much better position to revisit the Universe on a fully relativistic footing.

The focus of this chapter is an introduction to the study of metrics embodying the large-scale accelerated expansion of the Universe and the physics thereof, rather than a comprehensive presentation of modern cosmology, a truly vast subject. Our discussion of the foundations of observational cosmology, §10.7, for example, is largely descriptive. The reader desiring further or more technical information should consult the references in the bibliography.

10.1 A dynamic Newtonian universe

A plausible, if somewhat naïve, first model of the Universe might be one in which space is an ordinary static Euclidean geometry, and the galaxies fill up this space uniformly (on average) everywhere out to infinity. Putting aside the question of the origin of such a structure (let us say it has existed for all time) and the problem that the cumulative light received at any location would be infinite (“Olber’s paradox”—that is a bit more difficult to get around. Let us

Figure 10.1 In a static homogeneous Euclidean universe, an observer at the center of sphere 1 would calculate a different gravitational acceleration for the dot from that of an observer at the center of sphere 2: $a_1 \neq a_2$. But if we take into account the actual relative acceleration of the two observers, each considers the other to be in a *noninertial* frame, and the calculation becomes self-consistent with this noninertial “fictitious” force. (See the text.)



say that the galaxies turned on at some finite time in the past¹), the static Euclidean model is a nonstarter, because it is not even mathematically self-consistent.

Consider figure [10.1]. In the top diagram, we note that there are two observers, one at the center of sphere 1, the other at the center of sphere 2. Each calculates the expected acceleration at the location of the big black dot, which is a point on the surface of each of the spheres at a distance r_1 and r_2 from their respective centers. Our model universe is spherically symmetric about sphere 1, but it is also spherically symmetric about sphere 2! Hence, we have the following conundrum: The observer at the center of sphere 1 ignores the effect of the spherically symmetric mass exterior to the black point and concludes that the acceleration at the dot's location is

$$a_1 = \frac{GM(\text{within } r_1)}{r_1^2} = \frac{4\pi G\rho r_1}{3}, \quad (10.1)$$

directed toward the center of sphere 1. (Here ρ is the average uniform mass density of the Universe.) But the observer at the center of sphere 2 claims, by identical reasoning, that the acceleration must be $a_2 = 4\pi G\rho r_2/3$ directed toward the center of sphere 2. This is shown in the bottom diagram of figure [10.1]. Both cannot be correct.

But what if the Universe is dynamically active? Then, as Newtonian physicists, we must put in the gravitational acceleration in the form of a noninertial reference frame from the very start of the calculation. If the observers at the centers of spheres 1 and 2 are accelerating relative to each other, there is no reason to expect their separate calculations for the black dot acceleration to agree, because the observers are not part of the same inertial frame! Can we make this picture self-consistent somehow for any two r_1 and r_2 observers? Yes, provided that the Universe is allowed to expand (or contract) and Newtonian mechanics is applied self-consistently.

¹See W72, pp. 611-13.

Here is how it would work. The observer at the center of sphere 1 measures the acceleration of the black dot to be $-4\pi G\rho\mathbf{r}_1/3$, as above, with $\mathbf{r}_1$ pointing from the center of sphere 1 to the surface dot. The same sphere 1 observer finds that the acceleration of the center of sphere 2 is $-4\pi G\rho(\mathbf{r}_1 - \mathbf{r}_2)/3$, where $\mathbf{r}_2$ is the position vector oriented from the sphere 2 center toward the big dot. Defined this way, these particular $\mathbf{r}_1$ and $\mathbf{r}_2$ vectors are collinear. Thus, the sphere 1 observer finds that the acceleration of the big dot, *as measured by an observer moving in the (noninertial) center of sphere 2 frame* is the sphere 1 value of $-4\pi G\rho\mathbf{r}_1/3$ minus the acceleration of the observer at the center of sphere 2:

$$-\frac{4\pi G\rho\mathbf{r}_1}{3} - \frac{-4\pi G\rho(\mathbf{r}_1 - \mathbf{r}_2)}{3} = -\frac{4\pi G\rho\mathbf{r}_2}{3} \quad (10.2)$$

This is precisely the acceleration that the sphere 2 central observer would find self-consistently on their own. A Euclidean “linearly accelerating” universe is, in this sense, perfectly self-consistent. The need for expansion is a requirement of Newtonian dynamics, not, as originally thought at the time of its discovery, a mysterious effect of general relativity. One would think that the rate of expansion ought to be slowing, since this is what gravity does: an object thrown from the surface of Earth, in effect a little piece of Earth itself, slows down as it expands away from its parent body. This is certainly what astronomers assumed for 70 years. It turns out, however, that our Universe is more devious than that. There is some real mystery here beyond the Newtonian realm.

10.1.1 Equation of motion

A simple way to describe the internal acceleration of the Universe is to begin with the spatially homogeneous time-dependent relative expansion between two spatial locations. We may write separation between two arbitrary points separated by a distance $r(t)$ as

$$r(t) = R(t)l, \quad (10.3)$$

where l is a comoving coordinate that labels a “fixed” radial distance from us in space, in the sense of being fixed to the expanding space, like latitude and longitude would be on the surface of an inflating globe. It is natural to take l to have dimensions of length, in which case $R(t)$ is a dimensionless function of time alone, a scale factor that, via its time dependence, embodies the dynamic behavior of the Universe. The velocity $v = dr/dt \equiv \dot{r}$ of a comoving point expanding with space (i.e., at fixed l) is then

$$v(t) = \dot{R}l = (\dot{R}/R)r. \quad (10.4)$$

It is best to emphasize the vector character of this relationship:

$$\mathbf{v}(t) = (\dot{R}/R)\mathbf{r}, \quad (10.5)$$

where $\mathbf{r}$ is a standard vector pointing outward from our arbitrarily chosen origin. The acceleration is then

$$\mathbf{a}(t) = \frac{d\mathbf{v}}{dt} = (\ddot{R}/R)\mathbf{r}. \quad (10.6)$$

This kinematic relative acceleration between two points is, by Newtonian gravitation theory, given by equation (10.1):

$$\mathbf{a}(t) = -\frac{4\pi G\rho}{3}\mathbf{r}. \quad (10.7)$$

Hence,

$$-\frac{4\pi G\rho}{3} = \frac{\ddot{R}}{R}. \quad (10.8)$$

Notice that l has disappeared: this is an equation for R only, and this scale factor depends not on where you are, but on *when* you are. Next, we multiply this equation by $\dot{R}R$ and integrate, assuming that ordinary matter is conserved in an expanding comoving volume, i.e., ρR^3 is constant. We then obtain

$$\dot{R}^2 - \frac{8\pi G\rho R^2}{3} = 2E, \quad (10.9)$$

where E is an energy-like integration constant. In a simple, apparently naïve Euclidean-Newtonian approach, this would be our fundamental dynamic cosmological equation for the evolution of the Universe. Amazingly, providing that we are prepared to allow the mass density ρ to be upgraded to a full energy density (divided by c^2) that includes *all* contributions (in particular, radiation and vacuum energy), this innocent little equation turns out to be far more general: it is *exactly* correct in full relativity theory! We shall have more to say on this topic.

10.1.2 Cosmological redshift

The expansion of the Universe leads to a very important kinematic effect known as the cosmological redshift. Since the Universe is expanding, a traveling photon is constantly overtaking more distant observers that are moving away from the original source. If the photon has a measured wavelength λ at location r , then when the photon later passes an observer a small distance $dr = c dt$ away from r , who is moving at a relative velocity $\dot{R} dr/R$, this observer would measure a Doppler change in wavelength $d\lambda$ determined by equation (10.5):

$$\frac{d\lambda}{\lambda} = \frac{v}{c} = \frac{\dot{R}}{cR} dr = \frac{\dot{R}}{R} dt, \quad (10.10)$$

or, in other words

$$\frac{1}{\lambda} \frac{d\lambda}{dt} \equiv \frac{\dot{\lambda}}{\lambda} = \frac{\dot{R}}{R}. \quad (10.11)$$

Solving this simple differential equation for λ , we find that it is linearly proportional to R : it is as though the wavelength stretches out with the rest of the Universe. This is a very general kinematic result, a property of any cosmological model that is symmetrically expanding. (Even if there is spatial curvature, the “differential proper distance” is always $c dt$ for a traveling photon.)

This result means that the frequency of a photon goes *down* as $1/R$. But the frequency of a photon is proportional to its energy, hence to the radiation temperature T_γ which then also must be $\propto 1/R$. The energy density in radiation (fourth power in T_γ) decreases as $1/R^4$. Evidently, the total radiation energy in a comoving volume R^3 decreases as $1/R$: the quantity

we identify as energy is *not* conserved by the expansion of the Universe. By contrast, the *entropy* density in the radiation is proportional to T_γ^3 , and $(T_\gamma R)^3$ is therefore constant. The radiation entropy, which is just proportional to the number of photons, is conserved. Photons are neither created nor destroyed by the act of expansion. This makes perfect sense.

Where does the energy go? The radiation stress-energy tensor $T^{\mu\nu}$ is certainly conserved in the sense of $T^{\mu\nu}_{;\mu} = 0$, but there is still an exchange between the energy-like terms and the geometry, owing to the presence of the affine connection in the covariant derivative. These same affine connection terms arise even in the simplest form of the geodesic equation which makes contact with Newtonian dynamics, equation (3.27). There, we find it convenient to associate the affine contribution with “potential energy,” and keep energy conservation intact. But from a relativistic spacetime perspective, this is an example of the type of exchange embedded in the covariant formalism: the kinetic energy has gone into gravitational potential energy. Gravity *is* geometry.

We are free, and it is customary, to choose our coordinates in such a way that the current value of R is 1, with R becoming smaller as we go back in time. If a photon was emitted with a wavelength λ_e at some time t in the past, the wavelength we would now measure at time t_0 is formally given by

$$\lambda(t_0) = \lambda_0 = \lambda_e(1 + z), \quad (10.12)$$

where z , as defined by this equation, is known as the *redshift parameter*. Therefore,

$$\frac{\lambda_0}{\lambda_e} = 1 + z = \frac{R(t_0)}{R(t)} = \frac{1}{R(t)}. \quad (10.13)$$

The advantage of using z , as opposed to the more geometric quantity R , is that z is directly observed by astronomers in their instruments. But the two quantities are mathematically equivalent via the general equation $R = 1/(1 + z)$. In *any* spherically symmetric cosmological model, if you measure a redshift of 2, it has come from a time when the Universe was one-third its current size.

10.2 Cosmology models for the impatient

10.2.1 The large-scale spacetime metric

Although the Newtonian approach works rather well, let us remember that we really do need relativity in our study of the large-scale structure of the Universe.

First, we require a Riemannian metric structure to ensure that the speed of light is a proper universal constant c when traversing a dynamically evolving spacetime background. It is easy to see what the form of this metric must be in a simple model of an expanding Euclidean Universe. Symmetry demands that time must flow identically for all observers comoving with the universal expansion, and if g_{00} does not depend upon position, we can always choose our time coordinate so that g_{00} is a constant. A Euclidean space uniformly expanding everywhere—the metric practically leaps out of the page:

$$-c^2 d\tau^2 = -c^2 dt^2 + R^2(dx^2 + dy^2 + dz^2). \quad (10.14)$$

We have used here the usual (x, y, z) Cartesian coordinates, and $R = R(t)$ satisfies equation (10.9). Note that x, y, z are comoving with the expansion, with an embedded Euclidean



Figure 10.2 An ant crawling from the north pole to the equator at a constant physical velocity c on an expanding globe captures the essentials of a photon traveling through the spacetime of an expanding universe. With R a suitably normalized time-dependent radius, and dl the change in latitude angle, equation (10.15) is precisely obeyed. The ant may never reach the equator if $\dot{R} > 0$.

comoving spatial interval dl given by

$$dx^2 + dy^2 + dz^2 = dl^2.$$

In particular, a photon heading directly toward us (“the observer”) satisfies the equation

$$R \frac{dl}{dt} = -c, \quad (10.15)$$

where dl is now, by proper orientation of our axes, the change of radial comoving coordinate induced by the photon’s passage. A useful mechanical analog is that of an ant moving at a constant velocity (c for present purposes), crawling along the surface of an expanding globe from the pole to the equator (Fig. [10.2]). In this case, dl would correspond to the change in the ant’s latitude ($-d\theta$ in the standard colatitude angle).

Second, we must allow for the theoretical possibility of non-Euclidean spatial structure. As it happens, equation (10.14) really does appear to be the true form of the spacetime metric for our Universe. Space seems to be very nearly, or perhaps even precisely, Euclidean. But spatial flatness need not be the case even if we demand perfect symmetry, any more than the requirement of a perfectly symmetric two-dimensional surface demands a Euclidean plane. We could preserve our global *maximal spatial symmetry* and be in a mathematically curved space, like the surface of a sphere. Such a surface, in common with a flat plane, is symmetric about every point, but its spatial properties are clearly distorted relative to a plane. (E.g., the Pythagorean theorem does not hold, and the sum of the angles of a triangle on the surface exceeds 180° .) The nature of a two-dimensional spherical surface is readily grasped because we can easily embed it in three dimensions to form a mental image—just look at a globe. The surface is finite in area and is said to be *positively* curved. There is also a perfectly viable, flaring, *negatively* curved two-dimensional surface. A saddle begins to capture its essence, but not quite, because the curvature of a saddle is not uniform. A uniformly negatively curved two-dimensional surface cannot be smoothly embedded in three Euclidean dimensions (a result proved by Hilbert), so it is very hard to picture in your mind’s eye! There are positively and negatively curved *three-dimensional* spaces, as well, all perfectly symmetric, all logically possible alternative structures for the space of our Universe. They

just happen not to fit the observations very well. It is most fortunate for us that the real Universe seems also to be mathematically the simplest in its geometry. For now, we shall confine our attention to this expanding, familiar flat Euclidean space.

Third, we need relativity in the form of the Birkhoff theorem to properly justify neglecting exterior contributions from outside the arbitrarily chosen spheres we invoked in §10.1. A Newtonian description should not be applied to an infinite system, whereas nothing prevents us from applying Birkhoff's theorem to an unbounded symmetric spacetime.²

Fourth, at some point we shall require a relativistically valid argument to arrive at equation (10.9). Nothing in our Newtonian derivation hints at this level of generality. We shall return to this much more general calculation in §10.3.

Finally, we need relativity theory to relate the algebraic energy constant E to the geometry of our space. For now, we restrict ourselves to the case $E = 0$, which will turn out to be the only solution consistent with the adoption of a flat Euclidean spatial geometry, our own Universe.

10.2.2 The Einstein–de Sitter universe: A classic benchmark model

Consider equation (10.9) for the case $E = 0$ in the presence of ordinary matter and nothing else, which means that ρR^3 is a constant. Remember that we are free to choose coordinates in which $R = R_0 = 1$ at the present time $t = t_0$. We may then choose the constant ρR^3 to be equal to $\rho(t_0) = \rho_{M0}$, the mass density at the current epoch. Equation (10.9) becomes

$$R^{1/2} \dot{R} = \left(\frac{8\pi G \rho_{M0}}{3} \right)^{1/2}. \quad (10.16)$$

Integrating, we obtain

$$\frac{2}{3} R^{3/2} = \left(\frac{8\pi G \rho_{M0}}{3} \right)^{1/2} t, \quad (10.17)$$

where we set the integration constant to zero under the assumption that R was very small at early times. This leads to

$$R = \left(\frac{3H_0 t}{2} \right)^{2/3}, \quad (10.18)$$

where we have introduced an important structural constant of this so-called Einstein–de Sitter universe, H_0 , the value of $\dot{R}$ at the current time t_0 . This is known as the Hubble constant. In the example at hand,

$$H_0 = \dot{R}(t_0) = \left(\frac{8\pi G \rho_{M0}}{3} \right)^{1/2} \rightarrow \dot{R} = H_0 R^{-1/2}. \quad (10.19)$$

More generally, for any model universe, we define the Hubble *parameter* as the logarithmic derivative of $R(t)$:

$$H(t) = \frac{\dot{R}}{R} \quad (10.20)$$

²Strictly speaking, we should apply Birkhoff's theorem to an evacuated spherical shell, which we then allow to approach zero thickness.

for arbitrary time t . This definition holds independently of the normalization of R ; with $R(t_0) = 1$ the Hubble constant $H_0 = \dot{R}_0$, as seen in equation (10.19).

Exercise 10.2.1

Show that $H(t) = H_0(1+z)^{3/2}$ and $\rho(t) = 1/6\pi G t^2$ for our simple Einstein–de Sitter model $R = (t/t_0)^{2/3}$. Note that ρ depends only upon G and t .

In principle, we may observe the Hubble constant directly by measuring both the distances to nearby galaxies, as well as their redshifts, and then using equation (10.5) locally. As a practical astronomical problem, this is hardly straightforward! The task is difficult because we cannot be certain that the motion of nearby galaxies comes from cosmological expansion alone, isolated from local interactions. But astronomers have persevered, obtaining reliable distance measurements to ever more distant galaxies, and the result is that *the measured value of H_0 and the measured value of the density of ordinary matter ρ_{M0} do not satisfy equation (10.19) in our Universe, even allowing room for reasonable error*. There is simply not enough ordinary matter ρ_{M0} to account for the measured H_0 . However, equation (10.9) with $E = 0$ also seems to be well satisfied: the Universe appears to be very flat in its spatial geometry. The energy density of radiation in the contemporary Universe is only a tiny fraction of $3H_0^2/8\pi G$, so how is all this possible?

The answer is stunning. While the energy density of *ordinary* matter dominates over radiation, it is *not* the dominant contributor to the energy density of the Universe. There is now strong evidence of an additional energy density associated with the vacuum of space-time itself! This energy density, ρ_V , is the dominant energy density of the real Universe on cosmological scales, though not at present overwhelmingly so: ρ_V is about 69% of the total energy budget, whereas matter (the familiar baryons plus whatever dark matter is) comes in at about 31%. However, because ρ_V remains constant as the Universe expands, at later times vacuum energy will completely dominate the expansion. While ρ_M drops off as $1/R^3$, the constant ρ_V takes up the slack.

What does this imply for the later expansion of the Universe? With an effective vacuum Hubble parameter of

$$H_V \equiv \left(\frac{8\pi G \rho_V}{3} \right)^{1/2}, \quad (10.21)$$

equation (10.9) at later times takes the form

$$\dot{R} = H_V R \quad (10.22)$$

or

$$R \propto \exp(H_V t). \quad (10.23)$$

The Universe will expand exponentially. Rather than gravity slowly decelerating the expansion by the mutual attractive force of Newtonian gravity, the vacuum energy density will actively drive an ever more vigorous expansion via what amounts to a repulsive force. The converse of this is that an exponential universe would have been expanding less rapidly in

the past than in the present. It is this “less rapidly than expected” earlier history of cosmic expansion that has led to our current understanding of the remarkable dynamic behavior of our Universe. (The Nobel Prize in Physics was awarded to Perlmutter, Schmidt, and Riess in 2011 for their use of distant supernovas as a tool for unraveling the dynamics of the Universe from early to modern times.) We currently live, it so happens, in the epoch in which exponential expansion is beginning to take over from deceleration.

Let us return to the Einstein–de Sitter (E-dS) universe. Equations (10.13), (10.15), and (10.19) may be combined to answer the following question. If we measure a photon of redshift z , from what value of l did it originate? This is an important question because it provides the link between astronomical observations and the geometry of our Universe: $l(z)$ is a different function for different spacetime geometries. For the Einstein–de Sitter case $R(t) = (t/t_0)^{2/3} = (3H_0 t/2)^{2/3}$, and we may integrate equation (10.15) over the path of the photon from its emission at l at time t :

$$\int_0^l dl' = -c \int_{t_0}^t \frac{dt'}{R} = c \int_R^1 \frac{dR'}{\dot{R}' R'} = \frac{2c}{H_0} (1 - \sqrt{R}) = \frac{2c}{H_0} \left(1 - \frac{1}{\sqrt{1+z}}\right) \equiv l(z). \quad (10.24)$$

Note that as $z \rightarrow \infty$ (or $R \rightarrow 0$), $l \rightarrow 2c/H_0$, a constant. The most distant photons—and therefore the most distant regions of the Universe that may causally influence us today—come from $l = 2c/H_0$. This quantity is known as the *horizon*, denoted by l_H . Quite generally,

$$l_H = c \int_0^1 \frac{dR}{\dot{R}}, \quad (10.25)$$

with the cosmological model entering through the choice of $\dot{R}$. Beyond the horizon, we can detect—and be influenced by—nothing. The particular value of $l_H = 2c/H_0$ is associated with the Einstein–de Sitter model, but the existence of a horizon is a general feature of many cosmologies. On dimensional grounds it will generally be of order c/H_0 . Do not confuse this cosmological horizon, sometimes called a “particle horizon” with the *event horizon* of a black hole. The particle horizon is the distance over which cosmological-scale causality may be exerted. The event horizon is a boundary surface enveloping a black hole (a three-dimensional hypersurface when its extent through time is properly included), within which no causal contact with the outside world is possible.

Physical distances are given not by comoving coordinate l , but by Rl . Since $R = 1$ currently, $2c/H_0$ is the physical scale of the horizon today in an E-dS universe. There is nothing special about *now*, however. We could be doing this analysis at any time t , and the physical size of the horizon at time t in the E-dS model would then be $2c/H(t) = 2c/[H_0(1+z)^{3/2}]$. So, here is another interesting question. As a practical matter, the farthest back in time that we can see is to a redshift of about $z = 1100$. At higher redshifts, the Universe was completely opaque to photons. Just as in viewing the Sun we can see only down to its photosphere and not into the interior (at least not with photons), we can see in cosmological time only back to an era when the Universe itself was becoming similarly opaque. What would be the subtended *angular size* on the sky *today* of the physical horizon distance at a redshift of $z = 1100$? This is an important question, because we would not expect the Universe to be very smooth, or regular, or correlated in any way on angular scales larger than this. (The E-dS model doesn’t actually hold during the radiation-dominated phase, but its use in the calculation will serve to make our point; greater numerical accuracy doesn’t help

the problem!) The E-dS angular size of the horizon at a redshift z is given by the following expression:

$$\Theta_H(z) = \frac{2c}{H_0(1+z)^{3/2}} \times \frac{1}{R(z)l(z)} = \frac{2c}{H_0(1+z)^{1/2}l(z)} = \frac{1}{\sqrt{1+z}-1}. \quad (10.26)$$

The first equality sets the horizon angle equal to the actual physical size of the horizon at redshift z , divided by $d = R(z)l(z)$, the distance to redshift z at *precisely the same earlier time corresponding to redshift z* (not now!). This d is the relevant proper distance to the photon source at the moment of the radiation emission when the proper size was $2c/H(t)$. The Universe was a smaller place then, and it would be inconsistent to compute the angle $\Theta_H(z)$ using the *current* proper distance $R_0l(z) = l(z)$, with the *old* proper horizon size. If you plug in $z = 1100$ into equation (10.26) and convert Θ_H to degrees, you'll find $\Theta_H = 1.8^\circ$, a bit more than three times the diameter of the full moon. But the actual Universe looks very regular on much larger angular scales. Indeed, we know that the radiation temperature over the entire sky is unchanging to roughly a part in 10^4 . Even if our E-dS model is only crude, it highlights an important problem. Simply put, how does the Universe know about itself in such a globally coordinated manner, given that it takes signals traveling at the speed of light much longer than the age of the Universe to cross it? *What we've got here is a failure to communicate.* We will see later how modern cosmology can address this puzzle and facilitate communication.

The fact that the Universe was “on top of us” at early times has another surprising consequence. Assuming that the average physical size of a galaxy isn't changing very much with time, if we calculate the average *angular* size of a galaxy, we find that at low redshift, all is normal: the more distant galaxies appear to be smaller. But then, at progressively higher redshifts, the galaxies appear to be growing larger on the sky! Why? Because at large z , the Universe was, well, on top of us. The formula for the proper distance to redshift z is (for E-dS models)

$$l(z)R(z) = \frac{2c}{H_0(1+z)^{3/2}} (\sqrt{1+z}-1). \quad (10.27)$$

At low z , this is increasing linearly with z , which is intuitive: bigger redshift, more distant. But at large z this *declines* as $1/z$, since the Universe really was a smaller place.

Exercise 10.2.2

At what redshift would the average galaxy appear to be smallest in this model? (Answer: $z = 5/4$.)

10.2.3 The growth of density perturbations in an expanding universe

In an expanding universe, the density of nonrelativistic matter decreases as $1/R^3$. As the reservoir of material to form condensed objects is quickly diminishing in such a background flow, does gravity result in the sort of runaway collapse we normally think of when we

envision a galaxy forming? The Einstein–de Sitter model is a simple and revealing venue for examining this extremely important question. To determine the fate of a small overdensity or underdensity of nonrelativistic matter in an expanding universe is a problem that can be approached via an analysis of how small perturbations behave in a uniformly expanding background. Moreover, we can profitably approach this problem entirely in the Newtonian regime.

We require two simple fundamental equations. The first is simply the conservation of ordinary matter:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0, \quad (10.28)$$

where, as usual, ρ is the mass density and $\mathbf{v}$ is the velocity. The other is Newton's familiar equation of motion, which states that if a mass element is acted on by a gravitational force given by $-\rho dV \nabla \Phi$, where dV is a volume element comoving with the element and Φ is the associated potential function, the mass of fluid ρdV is accelerated by $-\nabla \Phi$:

$$\left[\frac{\partial}{\partial t} + (\mathbf{v} \cdot \nabla) \right] \mathbf{v} = -\nabla \Phi. \quad (10.29)$$

The acceleration measured relative to a fixed spacetime coordinate background means, of course, that the total time derivative must be used, $\partial_t + v_i \partial_i$ in index notation.

The background expansion of the universe is described by the usual Hubble law:

$$\mathbf{v} = \frac{\dot{R}}{R} \mathbf{r}. \quad (10.30)$$

Assuming spherical symmetry, the mass equation becomes

$$\frac{\partial \rho}{\partial t} + \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^3 \rho \frac{\dot{R}}{R} \right) = 0. \quad (10.31)$$

With the background density $\rho(t)$ independent of position, we then have,

$$\frac{d \ln \rho}{dt} + \frac{3\dot{R}}{R} = 0, \quad (10.32)$$

whence ρR^3 is a constant, as we know. As for the force equation, a direct calculation using equation (10.30) yields

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = \frac{\ddot{R}}{R} \mathbf{r}, \quad (10.33)$$

and, as the discussion of §10.1.1 shows that $\ddot{R}/R = -4\pi G\rho/3$, this is indeed the gravitational force from a sphere of radius r and density ρ . The Poisson equation

$$\nabla^2 \Phi = \frac{1}{r} \frac{\partial^2 (r\Phi)}{\partial r^2} = 4\pi G\rho \quad (10.34)$$

is likewise solved by our solution, with $\Phi = 2\pi G\rho r^2/3$.

We are interested in the fate of perturbative disturbances $\delta\rho$, $\delta\mathbf{v}$, and $\delta\Phi$ expressed as additions to this background:

$$\rho \rightarrow \rho + \delta\rho, \quad \mathbf{v} \rightarrow \mathbf{v} + \delta\mathbf{v}, \quad \Phi \rightarrow \Phi + \delta\Phi. \quad (10.35)$$

Using our fundamental equations, we replace our dynamic variables as shown, and because the δ -quantities are small, we retain them only through linear order, ignoring quadratic and higher-order terms. The mass equation is then

$$\frac{\partial \delta\rho}{\partial t} + \nabla \cdot (\rho \delta\mathbf{v}) + \nabla \cdot (\mathbf{v} \delta\rho) = 0. \quad (10.36)$$

Note that the gradient of ρ vanishes and that the equilibrium $\mathbf{v}$ satisfies $\nabla \cdot \mathbf{v} = -\partial_t \ln \rho$. It is then straightforward to show that the perturbed linearized mass conservation equation simplifies to

$$\left[\frac{\partial}{\partial t} + (\mathbf{v} \cdot \nabla) \right] \frac{\delta\rho}{\rho} + \nabla \cdot \delta\mathbf{v} = 0. \quad (10.37)$$

The linearized equation of motion

$$\left[\frac{\partial}{\partial t} + \mathbf{v} \cdot \nabla \right] \delta\mathbf{v} + (\delta\mathbf{v} \cdot \nabla) \mathbf{v} = -\nabla \delta\Phi \quad (10.38)$$

becomes

$$\left[\frac{\partial}{\partial t} + \mathbf{v} \cdot \nabla \right] \delta\mathbf{v} + \frac{\dot{R}}{R} \delta\mathbf{v} = -\nabla \delta\Phi. \quad (10.39)$$

Next, we change to coordinates which are comoving with the background Hubble expansion. This is straightforward. We define our spatial comoving coordinate $\mathbf{r}'$ by $\mathbf{r} = R(t')\mathbf{r}'$, and the comoving time remains unchanged: $t' = t$. The components of $\mathbf{r}'$ are x'_i . The partial derivative transformation is (summing, as usual, over repeated i)

$$\frac{\partial}{\partial t} = \frac{\partial}{\partial t'} + \frac{\partial x'_i}{\partial t} \frac{\partial}{\partial x'_i}, \quad \nabla = \frac{1}{R} \nabla', \quad \frac{\partial x'_i}{\partial t} = -\frac{x_i}{R^2} \dot{R} = -\frac{v_i}{R}, \quad (10.40)$$

so that

$$\frac{\partial}{\partial t} + (\mathbf{v} \cdot \nabla) = \frac{\partial}{\partial t'}, \quad (10.41)$$

which once again is a time derivative following a fluid element of the unperturbed background expansion. Our two equations for mass conservation and dynamics are therefore

$$\frac{\partial}{\partial t'} \frac{\delta\rho}{\rho} + \frac{1}{R} \nabla' \cdot \delta\mathbf{v} = 0, \quad (10.42)$$

$$\frac{\partial \delta\mathbf{v}}{\partial t'} + \frac{\dot{R}}{R} \delta\mathbf{v} = -\frac{1}{R} \nabla' \delta\Phi. \quad (10.43)$$

Taking ∇' of equation (10.43) and using equation (10.42) leads to

$$\frac{\partial^2}{\partial t'^2} \frac{\delta\rho}{\rho} + \frac{2\dot{R}}{R} \frac{\partial}{\partial t'} \frac{\delta\rho}{\rho} = \frac{1}{R^2} \nabla'^2 \delta\Phi = 4\pi G\rho \left(\frac{\delta\rho}{\rho} \right), \quad (10.44)$$

where the last equality is the linearized Poisson equation $\nabla^2 \delta\phi = 4\pi G\delta\rho$. For an Einstein-de Sitter universe, recall that $\rho = 1/6\pi Gt^2$ (see exercise 10.2.1) and $\dot{R}/R = 2/3t$, where we have now dropped the primes. Using the notation $\delta = \delta\rho/\rho$ and an overdot $\dot{\delta}$ for a time derivative, our differential equation for the growth of small perturbations in an Einstein de-Sitter universe takes on a very elegant form:

$$\ddot{\delta} + \frac{4}{3t} \dot{\delta} - \frac{2}{3t^2} \delta = 0. \quad (10.45)$$

This differential equation has two very simple linearly independent solutions, one in which δ decays as $1/t$, the other in which it grows as $t^{2/3}$. (Show!) The key point is that because of the background expansion, the perturbations display none of the rapid growth one usually finds against a static background. The growing power-law solution of the small perturbation is $t^{2/3}$, growing no faster than the universe itself is expanding, and certainly far from an exponential runaway. This is a pretty torpid tempo. For the musically inclined, think *adagio*.

How our Universe grew both its large- and small-scale structures has long been a mystery, one that remains far from well understood. To make structures grow in the barren gravitational soil of the Universe, one needs to start out with healthy-sized seeds. One also needs a great deal of dark matter, which is thought to consist of particles that interact only via their gravitational field. Alas, we must leave this fascinating topic untouched here.³ However, we revisit the question of how those first perturbations might have arisen, in §11.9.

10.3 The Friedmann-Robertson-Walker metric

10.3.1 Maximally symmetric 2-spaces

We begin with an important change of notation. Following convention, *the variable r will henceforth denote a comoving radial coordinate in expanding spatial geometries*. To avoid any confusion, we will no longer use l as a comoving coordinate.

The Friedmann-Robertson-Walker, or FRW, metric is the most general metric for a homogeneous isotropic cosmology. To understand its mathematical form, we start simply. The ordinary metric for a planar two-dimensional space (“2-space”) without curvature may be written in cylindrical coordinates as

$$ds^2 = d\varpi^2 + \varpi^2 d\phi^2, \quad (10.46)$$

where the radial ϖ and angular ϕ polar coordinates are related to ordinary Cartesian x - and y -coordinates by the familiar formulas

$$x = \varpi \cos \phi, \quad y = \varpi \sin \phi. \quad (10.47)$$

³See Ryden 2017 for further discussion.

As we have noted, this flat 2-space is not the most general globally symmetric 2-space possible. The space could, for example, be distorted like the 2-surface of a sphere and still retain the symmetry of the mathematical equivalence of every point. The metric for such a spherical surface is

$$ds^2 = a^2 d\theta^2 + a^2 \sin^2 \theta d\phi^2, \quad (10.48)$$

where a is the radius of the sphere. We know how to relate cylindrical polar and spherical coordinates: set $\varpi = a \sin \theta$. This may be viewed as a purely formal transformation of coordinates, but in our mind's eye we can picture θ in simple geometric terms as the colatitude angle measured downward from the z -axis, so that this ϖ is actually the same cylindrical radius ϖ seen in equation (10.46). But don't expect to get back to equation (10.46) from equation (10.48) via the simple coordinate change $\sin \theta = \varpi/a$. Instead, we find that the spherical surface metric becomes

$$ds^2 = \frac{d\varpi^2}{1 - \varpi^2/a^2} + \varpi^2 d\phi^2, \quad (10.49)$$

an entirely different space from the flat planar surface of equation (10.46). The space of equation (10.49) is the same original spherical surface space we started with by another name. The metric resembles equation (10.46) for a plane only in the limit $a \rightarrow \infty$. Changing coordinates does not change the geometry. In particular, a coordinate change cannot alter any curvature scalar, which remains invariant to coordinate transformation. Notice, as well, that ϖ is on large scales a “bad coordinate” in the sense that $\varpi = 0$ could correspond to either the north pole or the south pole of the sphere!

Let's stretch ourselves a bit and consider at a formal level the closely related two-dimensional metric

$$ds^2 = \frac{d\varpi^2}{1 + \varpi^2/a^2} + \varpi^2 d\phi^2, \quad (10.50)$$

which upon substitution of $\sinh \chi = \varpi/a$ reverts to

$$ds^2 = a^2 d\chi^2 + a^2 \sinh^2 \chi d\phi^2. \quad (10.51)$$

The fundamental symmetry properties of the metric are unaffected by the $\pm \varpi^2/a^2$ sign flip. The flip in denominator sign simply changes the sign of the constant curvature from positive (convex, think sphere) to negative (flaring, think saddle). This characteristic form of the radial metric tensor component, $g_{\varpi\varpi} = 1/(1 \pm \varpi^2/a^2)$, will reappear when we go from curved 2-space to curved 3-space.

Exercise 10.3.1

Show that the Ricci scalar R for the spherical metric, equation (10.48), is $-2/a^2$, and for the saddle metric equation (10.51) is $2/a^2$. These are constants of equal magnitudes across their respective surfaces, but with opposite signs. The underlying homogeneous and isotropic symmetry is the same in both cases.

10.3.2 Maximally symmetric 3-spaces

It is best to begin with the conclusion, which at this point ought not to surprise the reader. The most general form of metric tensor with a maximally symmetric (homogeneous and isotropic about every point) three-dimensional spatial structure takes the form

$$-c^2 d\tau^2 = -c^2 dt^2 + R^2(t) \left(\frac{dr^2}{1 - r^2/a^2} + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2 \right), \quad (10.52)$$

where we allow ourselves the abusive liberty of taking the curvature constant a^2 to be either positive or negative.

The derivation of the metric for the three-dimensional hypersurface of a four-dimensional sphere is not difficult. The surface of such a “4-ball” (Cartesian coordinates w, x, y, z) is given by

$$w^2 + x^2 + y^2 + z^2 = a^2 = \text{constant} > 0. \quad (10.53)$$

Thus, on the three-dimensional surface (“3-sphere”), a small change in w^2 is constrained to satisfy

$$d(w^2) = -d(x^2 + y^2 + z^2) \equiv -d(r^2), \text{ whence } dw = -rdr/w, \quad (10.54)$$

with $r^2 \equiv x^2 + y^2 + z^2$. Therefore,

$$(dw)^2 = \frac{r^2 (dr)^2}{w^2} = \frac{r^2 (dr)^2}{a^2 - r^2}. \quad (10.55)$$

Next, the line element in Cartesian 4-space is

$$ds^2 = dx^2 + dy^2 + dz^2 + dw^2 = dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2 + dw^2. \quad (10.56)$$

Adding together $(dr)^2$ and $(dw)^2$, as given by equation (10.55), immediately gives us the desired line element associated with the 3-sphere of a 4-ball, analogous to our expression for the ordinary spherical surface metric (10.48):

$$ds^2 = \frac{dr^2}{1 - r^2/a^2} + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2, \quad (10.57)$$

as seen in equation (10.52).

An alternative form for equation (10.57) is sometimes useful without the singular denominator. We set $r = a \sin \chi$. Then,

$$ds^2 = a^2 d\chi^2 + a^2 \sin^2 \chi d\theta^2 + a^2 \sin^2 \chi \sin^2 \theta d\phi^2. \quad (10.58)$$

Exercise 10.3.2

Show that the 3-sphere (surface) of a 4-ball with radius a has a “volume” $2\pi^2 a^3$. (Beware of factors of 2!)

Exercise 10.3.3

Would you care to hazard a guess as to what the line element of the 4-sphere of a 5-ball looks like in the form of either equation (10.57) or (10.58)?

What if we began not with the 3-sphere of a Euclidean 4-ball but with the three-dimensional surface of a four-dimensional hyperboloid,

$$x^2 + y^2 + z^2 - w^2 = -a^2 = \text{constant} < 0, \quad (10.59)$$

in a flat Minkowski-like space? The line element is

$$ds^2 = dx^2 + dy^2 + dz^2 - dw^2. \quad (10.60)$$

Why not? Special relativity is not the only venue for hyperbolic geometry.

An identical calculation constraining dw to stay within the three-dimensional surface now yields

$$(dw)^2 = \frac{r^2(dr)^2}{w^2} = \frac{r^2(dr)^2}{a^2 + r^2}. \quad (10.61)$$

The four-dimensional Minkowski metric, equation (10.60), constrained within our three-dimensional surface takes the form

$$ds^2 = dx^2 + dy^2 + dz^2 - dw^2 = \frac{dr^2}{1 + r^2/a^2} + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2. \quad (10.62)$$

This is the corresponding negatively curved (saddle-like) three-dimensional surface. With $r = a \sinh \chi$, its alternative form is

$$ds^2 = a^2 d\chi^2 + a^2 \sinh^2 \chi d\theta^2 + a^2 \sinh^2 \chi \sin^2 \theta d\phi^2. \quad (10.63)$$

For our Universe, a appears to be immeasurably large, but with χ compensatingly small, so that $a\chi$ is finite. In other words, space is neither positively nor negatively curved, at least not measurably so. Space is very close to flat.

It is customary in some treatments to set a to be a positive constant, use $r' = r/a$ as a rescaled radial variable, and then absorb the factor of a into the definition of what is meant by the scale factor $R(t)$. In effect, one introduces a new scale factor $R' = aR$. This eliminates a entirely from the metric. If you choose this route, however, you give up on the convenience of setting the current value of R' equal to 1. Dropping the prime on r' and R' , this general FRW metric, with a absorbed, we write

$$-c^2 d\tau^2 = -c^2 dt^2 + R^2(t) \left(\frac{dr^2}{1 - kr^2} + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2 \right), \quad (10.64)$$

where the constant k is $+1$ for a positively curved space, -1 for a negatively curved space, and 0 for a flat space. This particular line element with

$$g_{00} = -1, \quad g_{rr} = R^2(t)/(1 - kr^2), \quad g_{\theta\theta} = R^2(t)r^2, \quad g_{\phi\phi} = R^2(t)r^2 \sin^2 \theta, \quad (10.65)$$

is often what people mean when they write “the FRW metric.” The absence of a in this formulation can be mathematically convenient in performing calculations. In the author’s view, the form of equation (10.52) is more physically transparent.

Finally, be careful, lest the similarities lead to confusion. In equation (10.64), R has dimensions of length, and r is *dimensionless*. In equation (10.52), R is dimensionless, and r carries dimensions of length. For $k = 0$, we may stick to R being dimensionless and r carrying dimensions of length, as there is no need to “absorb a .” In all cases, r is a comoving coordinate.

Exercise 10.3.4

Show that by defining a new function of time $\eta = \eta(t)$, the metric in the form of equation (10.64) can be written in so-called conformal form as

$$-c^2 d\tau^2 = R^2(\eta) \left(-d\eta^2 + \frac{dr^2}{1 - kr^2} + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2 \right).$$

With this form of the metric, the scale factor R does not appear in the trajectories of the null geodesics. The new function $\eta(t)$ is known as conformal time. Give an explicit formula for $\eta(t)$, which you may leave in the form of an integral.

Exercise 10.3.5

- a) Consider a photon of 4-momentum p^μ traveling through a universe described by an FRW metric of the form of equation (10.64). Only p^0 and p^r components are present. Show that p_0 obeys the geodesic equation

$$\frac{dp_0}{dt} = \frac{1}{2} \frac{\partial g_{rr}}{\partial t} \frac{(p^r)^2}{p^0}.$$

(It may be helpful to start by treating the photons as ordinary massive particles and then take the limit of vanishing rest mass.) Solve this equation and show that its solution is $p_0 R = \text{constant}$ for all t ; the photon frequency decreases as $1/R$ with time. Note that this finding is independent of the FRW k -constant, and is fully consistent with our more elementary kinematic wavelength treatment of §10.1.2.

- b) Show that, solving for p^r ,

$$p^r = \pm \frac{p^0}{\sqrt{g_{rr}}},$$

and notice that p^r has a different (r, t) dependence from p^0 as the photon travels through the Universe. But quantum mechanically, p^0 is proportional to a frequency, and the radial 4-momentum p^r should just be proportional to a radial wavenumber, so the (r, t) dependencies should be the same for both. What error is being made in this reasoning? Fix it.

Exercise 10.3.6

Consider next an ordinary finite mass particle traveling through an FRW universe. Once again, you may choose your coordinates so that only the U^0 and U^r 4-velocity components are present. By solving the U_0 geodesic equation, show that

$$\sqrt{g_{rr}}RU^r = \text{constant}.$$

Interpret this result physically, and explain why it implies that an ensemble of such finite mass particles would be expanding adiabatically, once again not conserving energy, much as the cosmic background radiation.

A few final words on practical uses of the FRW metric. In this book, we shall use the metric in the form of equation (10.52). In flat Euclidean space, the integrated comoving distance to coordinate r is just r itself. (The *proper* distance is then found by multiplying by the scale factor $R(t)$ at the appropriate time.) Otherwise, the comoving radial distance associated with coordinate r is

$$\int_0^r \frac{dr'}{\sqrt{1 - r'^2/a^2}} = a \operatorname{asin}(r/a) \quad (a^2 > 0), \quad \text{or} \quad |a| \operatorname{asinh}(r/|a|) \quad (a^2 < 0).$$

(Recall that asin and asinh are the inverse sin and sinh function, respectively. Also, a is positive when $a^2 > 0$, and $|a|$ is the positive norm when $a^2 < 0$.)

By way of a specific example, let us rework equation (10.24) in our current notation for the case of positive curvature. The comoving coordinate of a photon arriving with redshift z when the scale factor was R may be expressed in terms of t , R , or z :

$$a \operatorname{asin}(r/a) = c \int_t^{t_0} \frac{dt'}{R(t')} = c \int_R^1 \frac{dR'}{\dot{R}'R'} = c \int_0^z \frac{dz'}{(1+z')\dot{R}(z')}.$$

Note that we have used $R = 1/(1+z)$, so that the current value of R is 1, which is appropriate for the form of the metric in equation (10.52). $\dot{R}$ is given by equation (10.9) quite generally, but we need to relate E and a . That requires the field equations. In §10.4, we show that $2E = -c^2/a^2$.

Let us next look at a flat, pure vacuum universe, but one with finite ρ_V and H_V so that we have something to work with! Then, $\dot{R} = H_V R$, and

$$r = c \int_R^1 \frac{dR'}{\dot{R}'R'} = \frac{c}{H_V} \int_R^1 \frac{dR'}{R'^2} = \frac{c}{H_V} \left(\frac{1}{R} - 1 \right) = \frac{cz}{H_V}. \quad (10.66)$$

How interesting. There is no horizon in such a universe. (Nor, for that matter, is there even a big bang.) The point is that as $z \rightarrow \infty$, $r \rightarrow \infty$. An early phase of exponential growth (“inflation”) has, in fact, been proposed as a means of avoiding the horizon problem. We’ve learned something very important already.

Next, the volume in a maximally symmetric space within redshift z_m is formally given by

$$V = 4\pi \int_0^r R^3 \frac{r'^2 dr'}{\sqrt{1 - r'^2/a^2}}, \quad (10.67)$$

a function of r and thus ultimately of redshift. But what do we use for R inside this integral? That depends on the question. If we are interested in the *current* net volume of, let us say, a prescribed group of photon sources, then $R = R_0$, a constant, and life is simple. If we are interested in the net volume occupied by the sources *at the time of the photon emission*, then we need first to find R as a function of r to do the integral, then to find $r(z_m)$ to get back to redshift z . That can be complicated! As an easier example, however, the current net volume V for sources out to a maximum redshift z_m with $a^2 > 0$ is ($R = R_0 = 1$)

$$V = 4\pi \int_0^r \frac{r'^2 dr'}{\sqrt{1 - r'^2/a^2}} = 2\pi \left(a^3 \arcsin \frac{r}{a} - a^2 r \sqrt{1 - \frac{r^2}{a^2}} \right). \quad (10.68)$$

You should be able to show that this reduces to the more familiar $4\pi r^3/3$ when a is large. In that case, for an E-dS universe the volume to redshift z is (show!)

$$V(z) = \frac{32\pi}{3} \left(\frac{c}{H_0} \right)^3 \left(1 - \frac{1}{\sqrt{1+z}} \right)^3. \quad (10.69)$$

10.4 Large-scale dynamics

10.4.1 The effect of a cosmological constant

We begin with the field equations *including the cosmological constant*, equation (5.67):

$$R_{\mu\nu} - \frac{g_{\mu\nu}}{2} R = -\frac{8\pi G}{c^4} T_{\mu\nu} + \Lambda g_{\mu\nu}. \quad (10.70)$$

Recall the stress energy tensor for a perfect fluid:

$$T_{\mu\nu} = P g_{\mu\nu} + (\rho + P/c^2) U_\mu U_\nu. \quad (10.71)$$

We may then arrange the right-side source term of equation (10.70) as follows:

$$-\frac{8\pi G}{c^4} T_{\mu\nu} + \Lambda g_{\mu\nu} = -\frac{8\pi G}{c^4} \left[\tilde{P} g_{\mu\nu} + \left(\tilde{\rho} + \frac{\tilde{P}}{c^2} \right) U_\mu U_\nu \right], \quad (10.72)$$

where

$$\tilde{P} = P - \frac{c^4 \Lambda}{8\pi G}, \quad \tilde{\rho} = \rho + \frac{c^2 \Lambda}{8\pi G}. \quad (10.73)$$

In other words, the effect of a cosmological constant is to leave the left side of the field equations untouched and to leave the right side of the field equations in the form of a stress

tensor for a perfect fluid, but with the density acquiring a constant additive term $c^2\Lambda/8\pi G$ and the pressure acquiring a constant term of the opposite sign, $-c^4\Lambda/8\pi G$!

This is simple, almost trivial, mathematics, but profound physics. The effect of a cosmological constant is as if the vacuum itself had an energy density $\mathcal{E}_V = \rho_V c^2 = c^4\Lambda/8\pi G$ and a pressure $P_V = -c^4\Lambda/8\pi G$. Does it make sense that the vacuum has a negative pressure, equal to its energy density but opposite in sign? Yes! If the vacuum volume expands by dV , the change in energy per unit volume of expansion is just $dE/dV \equiv \rho_V c^2$. By the first law of thermodynamics, this must be $-P_V$. Yet more fundamentally, if we recall the form of the stress-energy tensor of the vacuum, *but without assuming* $P_V = -\rho_V c^2$, then

$$T_{\mu\nu}(V) = P_V g_{\mu\nu} + (\rho_V + P_V/c^2) U_\mu U_\nu, \quad (10.74)$$

and the last group of terms would change the form of the vacuum stress energy going from one constant-velocity observer to another. In other words, an observer could tell if they were moving relative to the vacuum. This is absolutely forbidden. The vacuum stress must always be proportional to $g_{\mu\nu}$, and to $g_{\mu\nu}$ alone. The only way this can occur is if $P_V = -\rho_V c^2$.

An early general relativity advocate, Sir Arthur Eddington, was particularly partial to a cosmological constant and famously commented that “To drop the cosmical constant would be to knock the bottom out of space.”⁴ At the time, this was probably viewed as Eddington speaking in his customary curmudgeon mode. Today, his insight seems quite prescient. Contemporary physicists think less in terms of a cosmological constant, placing greater conceptual emphasis instead on the notion of a vacuum energy density. Many questions then arise. What is the reason for its existence? Why does it have the value that it does? If ρ_V is not strictly constant, there must be new physics to be discovered. Are the best observational data supportive of a truly constant value for ρ_V ? The value of ρ_V probably emerged from the same type of “cancellation process,” or renormalization, that has somehow produced finite values for the masses for the fundamental particles. How can we calculate or even estimate this? These are some of the most important and difficult questions in physics today.

Moving forward, we must now place these fascinating issues to the side and continue our development of large-scale models of the Universe. We do so without the formal appearance of a cosmological constant in our equations, but with the understanding that we may add the appropriate contributions to the density ρ and pressure P to account precisely for the effects of Λ .

10.4.2 Formal analysis

Let us recall the field equations written in terms of the source function $S_{\mu\nu} = T_{\mu\nu} - g_{\mu\nu}T/2$,

$$R_{\mu\nu} = -\frac{8\pi G}{c^4} S_{\mu\nu}, \quad (10.75)$$

and the FRW metric in the compact form of equation (10.65). We shall lose no generality in doing so, as we may always set $k = 1/a^2$ at the end of the calculation without invalidating any

⁴Eddington, A. S. 1933, *The Expanding Universe* (Cambridge Univ.: Cambridge).

step along the way. In comoving coordinates, the only component of the covariant 4-velocity U_μ that does not vanish is $U_0 = -c$ (from the relations $g_{00}(U^0)^2 = -c^2$ and $U_0 = g_{00}U^0$). The nonvanishing components of $T_{\mu\nu}$ are then

$$T_{00} = \rho c^2, \quad T_{ij} = P g_{ij}. \quad (10.76)$$

This means

$$T^0_0 = -\rho c^2, \quad T^i_j = \delta^i_j P, \quad T = T^\mu_\mu = -\rho c^2 + 3P. \quad (10.77)$$

We shall also need

$$S_{00} = T_{00} - \frac{g_{00}}{2} T = \frac{1}{2}(\rho c^2 + 3P), \quad S_{ij} = \frac{1}{2}g_{ij}(\rho c^2 - P). \quad (10.78)$$

To calculate R_{00} , we begin with the Ricci tensor from equation (6.7):

$$R_{\mu\kappa} = \frac{1}{2} \frac{\partial^2 \ln g}{\partial x^\kappa \partial x^\mu} - \frac{\partial \Gamma_{\mu\kappa}^\lambda}{\partial x^\lambda} + \Gamma_{\mu\lambda}^\eta \Gamma_{\kappa\eta}^\lambda - \frac{\Gamma_{\mu\kappa}^\eta}{2} \frac{\partial \ln g}{\partial x^\eta},$$

where g is the absolute value of the determinant of the metric tensor $g_{\mu\nu}$ given by equation (10.65). Defining

$$f(r) = 1/(1 - kr^2),$$

we have

$$g = R^6(t) r^4 f(r) \sin^2 \theta. \quad (10.79)$$

For diagonal metrics, $\Gamma_{ab}^a = \partial_b(\ln |g_{aa}|)/2$, and $\Gamma_{aa}^b = -(\partial_b g_{aa})/2g_{bb}$ (no sum on a). Therefore, $\Gamma_{00}^\lambda = 0$, and our expression for R_{00} simplifies to

$$R_{00} = \frac{1}{2} \frac{\partial^2 \ln g}{\partial x^0 \partial x^0} + \Gamma_{0\lambda}^\eta \Gamma_{0\eta}^\lambda = \frac{1}{2} \frac{\partial^2 \ln g}{\partial x^0 \partial x^0} + \Gamma_{0r}^r \Gamma_{0r}^r + \Gamma_{0\theta}^\theta \Gamma_{0\theta}^\theta + \Gamma_{0\phi}^\phi \Gamma_{0\phi}^\phi. \quad (10.80)$$

With

$$\frac{1}{2} \frac{\partial^2 \ln g}{\partial x^0 \partial x^0} = \frac{3\ddot{R}}{Rc^2} - \frac{3\dot{R}^2}{R^2 c^2}, \quad \Gamma_{0r}^r = \Gamma_{0\theta}^\theta = \Gamma_{0\phi}^\phi = \frac{\dot{R}}{Rc}, \quad (10.81)$$

the 00 component of equation (10.75) becomes

$$\ddot{R} = -\frac{4\pi GR}{3} \left(\rho + \frac{3P}{c^2} \right), \quad (10.82)$$

which differs from our Newtonian equation (10.8) only by the additional term, seemingly very small, of $3P/c^2$ as an additional source of gravitation. However, during the time the Universe was dominated by radiation, this term was important and even now it turns out to be not only important but *negative* as well! During the so-called inflationary phase, $3P/c^2$ was hugely important. We will have much more to say about all this later in this chapter.

The rr -component of the field equations is a bit more involved. With $f' \equiv df/dr$, we prepare a working table in advance of all the results we will need:

$$R_{rr} = \frac{1}{2} \frac{\partial^2 \ln g}{\partial r^2} - \frac{\partial \Gamma_{rr}^\lambda}{\partial x^\lambda} + \Gamma_{r\lambda}^\eta \Gamma_{r\eta}^\lambda - \frac{1}{2} \Gamma_{rr}^\eta \frac{\partial \ln g}{\partial x^\eta}. \quad (10.83)$$

$$\Gamma_{rr}^r = \frac{1}{2g_{rr}} \frac{\partial g_{rr}}{\partial r} = \frac{f'}{2f}, \quad \Gamma_{rr}^0 = -\frac{1}{2cg_{00}} \frac{\partial g_{rr}}{\partial t} = \frac{fR\dot{R}}{c}, \quad \Gamma_{0r}^r = \frac{1}{2cg_{rr}} \frac{\partial g_{rr}}{\partial t} = \frac{\dot{R}}{Rc}. \quad (10.84)$$

$$\Gamma_{\phi r}^\phi = \frac{1}{2g_{\phi\phi}} \frac{\partial g_{\phi\phi}}{\partial r} = \Gamma_{\theta r}^\theta = \frac{1}{2g_{\theta\theta}} \frac{\partial g_{\theta\theta}}{\partial r} = \frac{1}{r}. \quad (10.85)$$

$$\frac{\partial \ln |g|}{\partial r} = \frac{f'}{f} + \frac{4}{r}, \quad \frac{1}{2} \frac{\partial^2 \ln |g|}{\partial r^2} = \frac{1}{2} \frac{f''}{f} - \frac{1}{2} \frac{(f')^2}{f^2} - \frac{2}{r^2}, \quad \frac{\partial \Gamma_{rr}^0}{\partial x^0} = \frac{1}{c^2} (f\dot{R}^2 + fR\ddot{R}). \quad (10.86)$$

$$\frac{\partial \ln |g|}{\partial t} = \frac{6\dot{R}}{R}, \quad \frac{\partial \Gamma_{rr}^r}{\partial r} = \frac{f''}{2f} - \frac{(f')^2}{2f^2}. \quad (10.87)$$

Putting it all together, we have

$$\begin{aligned} R_{rr} &= \frac{1}{2} \frac{\partial^2 \ln |g|}{\partial r^2} - \frac{\partial \Gamma_{rr}^r}{\partial r} - \frac{\partial \Gamma_{rr}^0}{\partial x^0} + (\Gamma_{rr}^r)^2 + (\Gamma_{\theta r}^\theta)^2 + (\Gamma_{\phi r}^\phi)^2 \\ &\quad + 2\Gamma_{rr}^0 \Gamma_{r0}^r - \frac{\Gamma_{rr}^r}{2} \frac{\partial \ln |g|}{\partial r} - \frac{\Gamma_{rr}^0}{2c} \frac{\partial \ln |g|}{\partial t} \\ &= \frac{f''}{2f} - \frac{(f')^2}{2f^2} - \frac{2}{r^2} - \left[\frac{f''}{2f} - \frac{(f')^2}{2f^2} \right] - \frac{1}{c^2} (f\dot{R}^2 + fR\ddot{R}) + \frac{(f')^2}{4f^2} \\ &\quad + \frac{2}{r^2} + \frac{2f\dot{R}^2}{c^2} - \frac{f'}{4f} \left(\frac{f'}{f} + \frac{4}{r} \right) - \frac{3f\dot{R}^2}{c^2} \end{aligned}$$

Thus, with $f'/f = 2krf$,

$$R_{rr} = -\frac{2f\dot{R}^2}{c^2} - \frac{fR\ddot{R}}{c^2} - 2kf = -\frac{8\pi G}{c^4} S_{rr} = -\frac{4\pi G}{c^4} g_{rr}(\rho c^2 - P) = -\frac{4\pi R^2 Gf}{c^4}(\rho c^2 - P),$$

or

$$2\dot{R}^2 + R\ddot{R} + 2kc^2 = 4\pi GR^2(\rho - P/c^2). \quad (10.88)$$

Notice that r has disappeared, as it must. (Why must it?) Eliminating $\ddot{R}$ from equation (10.88) via equation (10.82) and simplifying the result leads to

$$\dot{R}^2 - \frac{8\pi G\rho R^2}{3} = -kc^2. \quad (10.89)$$

This is exactly the Newtonian equation (10.9) with the constant $2E$ identified with $-kc^2$. *But be careful.* The Newtonian version of equation (10.9) was formulated with R dimensionless. If we use the standard FRW convention that k is either 1, 0, or -1 , then recall that R has been rescaled to have dimensions of length in equation (10.89). To recover the Newtonian

form is just a matter of replacing k with $1/a^2$ with dimensions of an inverse length squared. We then have $2E = -c^2/a^2$, sharing the same dimension of $1/\text{time}^2$. Remember that we must allow $a^2 < 0$ in the case of negative curvature. The important point is that we now know that equation (10.9) is valid in full general relativity. The dynamic Newtonian energy constant is therefore to be identified with the geometric curvature of the space!

There is one final item to consider in our analysis. We have not yet made explicit use of the equation for the conservation of energy-momentum, based on equation (4.74) and the Bianchi identities:

$$T^{\mu\nu}_{;\mu} = \frac{1}{\sqrt{g}} \frac{\partial(\sqrt{g}T^{\mu\nu})}{\partial x^\mu} + \Gamma^\nu_{\mu\lambda} T^{\mu\lambda} = 0. \quad (10.90)$$

This constraint is, of course, already built into the field equations themselves, and so in this sense adds no new information to our problem. But we may ask whether use of this equation from the start might have saved us some labor in getting to equation (10.89); it was a lengthy derivation after all. The answer is an interesting YES and NO.

The $\nu = 0$ component of equation (10.90) reads

$$T^{\mu 0}_{;\mu} = \frac{1}{\sqrt{g}} \frac{\partial(\sqrt{g}T^{\mu 0})}{\partial x^\mu} + \Gamma^0_{\mu\lambda} T^{\mu\lambda} = 0. \quad (10.91)$$

Only affine connections of the form Γ^0_{ii} (spatial index i , no sum) are present, and with

$$\Gamma^0_{ii} = -\frac{1}{2g_{00}} \frac{\partial g_{ii}}{\partial x^0} = \frac{1}{2} \frac{\partial g_{ii}}{\partial x^0}, \quad T^{00} = \rho c^2, \quad T^{ii} = P g^{ii} = \frac{P}{g_{ii}}, \quad (\text{NO } i \text{ SUMS})$$

equation (10.91) becomes

$$\frac{1}{R^3} \frac{\partial(R^3 \rho c^2)}{\partial t} + 3P \frac{\dot{R}}{R} = 0 \rightarrow \dot{\rho} + 3 \left(\rho + \frac{P}{c^2} \right) \frac{\dot{R}}{R} = 0. \quad (10.92)$$

This is simply the relativistic form of the first law of thermodynamics, $dE = -PdV$, which reverts to mass conservation in the nonrelativistic limit.

Exercise 10.4.1

Justify the last statement, and verify that a pure vacuum energy density universe trivially satisfies equation (10.92).

If we now substitute equation (10.92) for P in equation (10.82) and do a little rearranging, we arrive at the result

$$\frac{d}{dt} (\dot{R}^2) = \frac{8\pi G}{3} \frac{d}{dt} (\rho R^2), \quad (10.93)$$

which in turn immediately integrates to equation (10.89)! This is surely a much more efficient route to equation (10.89) . . . except that we cannot relate the integration constant that emerges from equation (10.93) to the spatial curvature constant k . True, we have a faster route to our final equation, but we have only dynamic content. Without explicitly

considering the Ricci R_{rr} component, we lose the geometric connection between Newtonian E (an untethered integration constant) and $-kc^2$ (a geometric curvature constant). We therefore have *consistency* between the energy conservation and Ricci approaches, but not true *equivalence*—a subtle and interesting distinction.

Our final comment is that the $3P\dot{R}/R$ contribution on the left of equation (10.92) is geometric in origin, stemming from an affine connection term. Absent this term, energy would be strictly conserved. This is another example of the interplay between energy and geometry. There can be no geometric expansion without some $T^{\mu\nu}$ to draw upon: an FRW universe with no ρ or P is a Minkowski universe, despite the presence of an ever-increasing FRW $R(t)$ scale factor. The next exercise illustrates this coordinate fallacy explicitly.

Exercise 10.4.2

- a) Show that for a classical vacuum $\rho = 0$ universe, the solution to equation (10.89) is $R = ct$, and therefore the FRW metric takes the form

$$-c^2 d\tau^2 = -c^2 dt^2 + \frac{c^2 t^2 dr^2}{1 + r^2} + c^2 t^2 r^2 (d\theta^2 + \sin^2 \theta d\phi^2).$$

This may *look* like an expanding universe, but it must just be Minkowski spacetime!

- b) Find the functions $s(r, t)$ and $T(r, t)$ such that

$$-c^2 d\tau^2 = -c^2 dT^2 + ds^2 + s^2 (d\theta^2 + \sin^2 \theta d\phi^2),$$

which we recognize as Minkowski spacetime. Note that $s = rct$. Justify this simple assertion. By further demanding that the coefficient of $c^2 dT^2$ be -1 , show that

$$T = t\sqrt{1 + r^2}.$$

- c) Verify in detail that the coefficient of $dTds$ vanishes, and the coefficient of ds^2 is unity under this transformation, thereby completing the coordinate transformation of the line element. (You may find it helpful to calculate in advance $(\partial t/\partial s)_T$ and $(\partial r/\partial s)_T$ in terms of r and s , and $(\partial r/\partial T)_s$ in terms of $(\partial t/\partial T)_s$.) Once again, we see how easy it is to be misled by coordinates!

10.4.3 The Ω_0 parameter

We begin with the dynamic cosmological equation in its “ $2E$ ” form, with no loss of generality,

$$\dot{R}^2 - \frac{8\pi G\rho R^2}{3} = 2E, \quad (10.94)$$

where E is simply an energy integration constant. This may be determined, in terms of directly observable quantities, by using the convention that $R = 1$ at the current time t_0 .

We write $2E$ in terms of H_0 , the Hubble constant $\dot{R}_0$, and $\rho_0 c^2$, the current average value of the total energy density of the Universe:

$$H_0^2 \left(1 - \frac{8\pi G \rho_0}{3H_0^2} \right) \equiv H_0^2 (1 - \Omega_0) = 2E = -\frac{c^2}{a^2}, \quad (10.95)$$

where we have introduced the parameter Ω_0 ,

$$\Omega_0 = \frac{8\pi G \rho_0}{3H_0^2}. \quad (10.96)$$

The Universe is therefore positively curved (closed) or negatively curved (open) according to whether the measured value of Ω_0 is larger or smaller than unity. Defining the critical mass density ρ_c by

$$\rho_c = \frac{3H_0^2}{8\pi G}, \quad (10.97)$$

we see that the critical *energy* density in the Universe is then $\rho_c c^2$. The parameter $\Omega(t)$ is defined just as Ω_0 , with $\rho(t)$ and $H(t)$ replacing their current t_0 values.

A puzzle associated with classical cosmologies that were based on the assumption that the Universe is composed of standard matter was that the measured value of Ω_0 seemed to be about 0.05. This is disquieting, because the two terms on the left side of equation (10.94) are then neither very close nor vastly different from each other—just different. But over the lifetime of the Universe, this interval of being “just different” is a mere fleeting instant of time. In the early Universe, the two terms were both large and remained very close in magnitude, as in an Einstein–de Sitter model. Thereafter, the $\dot{R}$ term dominates in a continuously expanding Universe, with $\Omega(t) \ll 1$ over almost all time. So why are we privileged to live in a very special epoch in which $\Omega(t)$ just happens to be in transition from unity to minuscule?

It is now understood that the constituent components of our Universe involve the energy density of the vacuum. The currently best measured value of H_0 (in standard astronomical units) is $67.6 \text{ km s}^{-1} \text{ Mpc}^{-1}$,⁵ or $H_0 \simeq 2.2 \times 10^{-18} \text{ s}^{-1}$. This number gives a critical density of $8.6 \times 10^{-27} \text{ kg}$ (about 5 hydrogen atoms) per cubic meter, far greater than what is actually observed in standard baryonic matter⁶. The best so-called “concordance models” all point to an $\Omega_0 = 1$, $E = 0$ Universe, but this requires $\sim 70\%$ of $\rho_0 c^2$ to come from the vacuum! About 25% comes from so-called dark matter, which is matter that is not luminous but whose presence is inferred from its gravitational effects, and as earlier measurements had indicated, only about 5% comes from ordinary baryonic matter in the form of gas and stars. We will henceforth denote the current vacuum contribution to Ω_0 as Ω_{V0} and the matter contribution as Ω_{M0} . In addition, there is a contribution from radiation,

⁵Recall that $1 \text{ Mpc} = 3.085678 \times 10^{19} \text{ km}$. Stars are typically separated by 1 pc in galaxies, galaxies from one another by about 1 Mpc .

⁶The absolute value of the *pressure* in the vacuum is some $5.4 \times 10^{-10} \text{ Pa}$ (for pascal, the SI pressure unit). This would be an ultravacuum pressure in a terrestrial lab but is rather substantial by astronomical standards: in magnitude, it is roughly the ordinary gas pressure in the core of a molecular cloud in the interstellar medium.

$\Omega_{\gamma 0}$. While it is quite negligible now, in the early universe ($z \geq 5500$) it was completely dominant. The Universe went first through a radiation-dominated phase, followed by a matter-dominated phase, and then at about a redshift of 0.3, it started to switch to its current vacuum-dominated phase. This latter transition is ongoing.

Exercise 10.4.3

A cluster of galaxies contains about 10^3 galaxies. With a contribution from dark matter, this amounts to about $10^{15} M_{\odot}$. With ρ_V taken to be $0.7\rho_c$, at what distance r from the cluster center would the average matter density within r equal that of the equivalent vacuum mass density? You may assume this distance lies outside the cluster. This gives a rough idea of the length scale over which the vacuum contribution becomes important for large-scale dynamics. (Answer: about 14 Mpc.)

While an $\Omega = 1$ Universe remains an $\Omega = 1$ Universe throughout all time, the relative contribution from each of its constituents changes, and the question of fine-tuning has not quite disappeared. We now just happen to be in the extended neighborhood of an inflection point where Einstein–de Sitter behavior is giving rise to exponential growth. It is not obvious what, if anything, to make of that.

10.5 The classic, matter-dominated universes

Matter-dominated models of the Universe dominated cosmological thought throughout the twentieth century. If the matter density is high, these universes all have a closed geometry and recollapse after a finite time. By contrast, low-density, open-geometry models expand forever, with the scale factor eventually coasting along linearly with time. A critical density divides these two cases. In this unique case the geometry is flat (Euclidean) and the expansion rate is ever diminishing but never halting: the scale factor is proportional to $t^{2/3}$ at all times t . This is the Einstein–de Sitter solution. All three types of solution may be studied analytically.

We therefore now return to the classic era of cosmology theory, when the idea that the Universe was driven by “the energy density of the vacuum” was not something that serious-minded astrophysicists dwelt upon. The expansion dynamics of the Universe was naturally thought to be dominated by matter, perhaps including dark matter, after a brief early phase when it was radiation dominated. Nonrelativistic matter obeys the constraint that $\rho(t)R^3(t)$ remains constant with time. With $R_0 = 1$, the constant in question must be the current mass density ρ_{M0} . The dynamic equation of the Universe may then be written as

$$\dot{R}^2 = 2E + \frac{8\pi G\rho R^2}{3} = H_0^2 \left(1 - \Omega_{M0} + \frac{\Omega_{M0}}{R} \right), \quad (10.98)$$

or

$$\int_0^R \frac{dR'}{\sqrt{1 - \Omega_{M0} + (\Omega_{M0}/R')}} = \int_0^R \frac{R'^{1/2} dR'}{\sqrt{(1 - \Omega_{M0})R' + \Omega_{M0}}} = H_0 t. \quad (10.99)$$

The nature of the integral depends upon whether Ω_{M0} is less than, equal to, or greater than 1. The case $\Omega_{M0} = 1$ is simple and leads immediately to

$$R = \left(\frac{3H_0 t}{2} \right)^{2/3}, \quad (10.100)$$

our Einstein–de Sitter solution, equation (10.18). According to equation (10.94) with $E = 0$, the average density is then given explicitly by

$$\rho = \frac{\dot{R}^2}{R^2} \frac{3}{8\pi G} = \frac{1}{6\pi G t^2}, \quad (10.101)$$

as we found in exercise 10.2.1.

We consider next the case, $\Omega_{M0} > 1$, that of a closed Universe. Then, we may write equation (10.99) as

$$\int_0^R \frac{R'^{1/2} dR'}{\sqrt{1 - (1 - \Omega_{M0}^{-1})R'}} = \Omega_{M0}^{1/2} H_0 t. \quad (10.102)$$

We set

$$(1 - \Omega_{M0}^{-1})R' = \sin^2 \theta'. \quad (10.103)$$

Then equation (10.102) becomes

$$\frac{2}{(1 - \Omega_{M0}^{-1})^{3/2}} \int_0^\theta \sin^2 \theta' d\theta' = \frac{1}{(1 - \Omega_{M0}^{-1})^{3/2}} \int_0^\theta (1 - \cos 2\theta') d\theta' = \Omega_{M0}^{1/2} H_0 t, \quad (10.104)$$

or

$$\frac{2\theta - \sin 2\theta}{2(1 - \Omega_{M0}^{-1})^{3/2}} = \Omega_{M0}^{1/2} H_0 t. \quad (10.105)$$

With $\eta = 2\theta$, our final solution is in the form of a parameterization:

$$R = \frac{1 - \cos \eta}{2(1 - \Omega_{M0}^{-1})}, \quad H_0 t = \frac{\eta - \sin \eta}{2\sqrt{\Omega_{M0}(1 - \Omega_{M0}^{-1})^3}}. \quad (10.106)$$

Some readers may recognize these equations for R and $H_0 t$ as the parameterization of a cycloid, which is the path taken by a fixed point on the circumference of a wheel as the wheel rolls forward. (See the curve labeled “Closed” in figure [10.3].) Precisely analogous expressions emerge for the case $\Omega_{M0} < 1$, which are easily verified from equation (10.99):

$$R = \frac{\cosh \eta - 1}{2(\Omega_{M0}^{-1} - 1)}, \quad H_0 t = \frac{\sinh \eta - \eta}{2\sqrt{\Omega_{M0}(\Omega_{M0}^{-1} - 1)^3}}. \quad (10.107)$$

This is the solution for negative curvature, an open Universe.

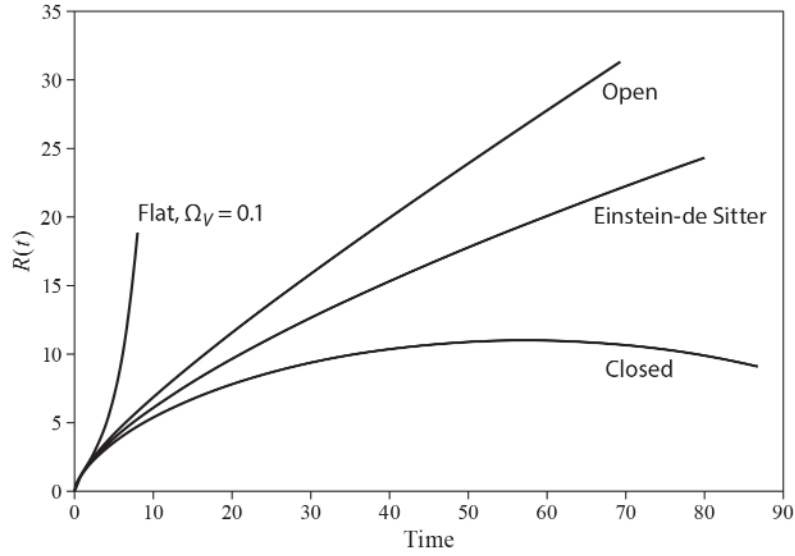


Figure 10.3 $R(t)$ versus time, units of $\Omega_M^{-1/2} H_0^{-1}$, for four model universes. Closed is equation (10.106); Einstein-de Sitter, equation (10.100); Open, equation (10.107). For Closed, $\Omega_M = 1.1$; for Einstein-de Sitter, $\Omega_M = 1$; for Open, $\Omega_M = 0.9$. The curve labeled “Flat, $\Omega_V = 0.1$,” equation (10.115), includes a cosmological constant with a vacuum contribution of $\Omega_V = 0.1$, so that $\Omega_M + \Omega_V = 1$, and the spatial geometry is Euclidean.

Exercise 10.5.1

Show that for either equation (10.106) or (10.107),

$$\frac{d\eta}{dt} = \frac{c}{|a|R},$$

where $|a|$ denotes a modulus. How is this η related to the conformal time η of Exercise 10.3.4?

Exercise 10.5.2

Show that the equations analogous to (10.100), (10.106), and (10.107) for the FRW metric in the form of equation (10.64) are, respectively,

$$R = \omega^{1/3} \left(\frac{3t}{2} \right)^{2/3} \quad (\text{Einstein-de Sitter}),$$

$$R = \frac{\omega}{c^2} \left(\frac{1 - \cos \eta}{2} \right), \quad ct = \frac{\omega}{c^2} \left(\frac{\eta - \sin \eta}{2} \right) \quad (\text{closed}),$$

$$R = \frac{\omega}{c^2} \left(\frac{\cosh \eta - 1}{2} \right), \quad ct = \frac{\omega}{c^2} \left(\frac{\sinh \eta - \eta}{2} \right) \quad (\text{open}),$$

where $\omega = 8\pi G\rho_{M0}R_0^3/3$. Note that $c \, dt = R \, d\eta$ for the last two cases.

Exercise 10.5.3

Show that as $\eta \rightarrow 0$, both equations (10.106) and (10.107) reduce to $R = \Omega_{M0}^{1/3} [3H_0 t/2]^{2/3}$ (identical, of course, with equation (10.100) for $\Omega_{M0} = 1$), and at late times equation (10.107) becomes the “coasting” solution, $R = \sqrt{1 - \Omega_{M0}} H_0 t$. This means that a plot of all possible solutions of $R(t)$ versus $\Omega_{M0} H_0 t$ would converge to exactly same solution at early times, regardless of the value of Ω_{M0} . Figure (10.3) shows this behavior quite clearly.

What if $\Omega_{M0} = 0$? Then $R = H_0 t$ for all times. But the expansion is only apparent; see exercise 10.4.2.

10.6 Our Universe

Prologue

Throughout most of the twentieth century, the goal of cosmology appeared to be to decide which of the three standard matter-dominated model scenarios actually holds: do we live in an open, closed, or critical universe? Solutions with a cosmological constant were relegated to the realm of slightly disreputable speculation, under the rather shameful rubric “Alternative Cosmologies” in the textbooks. All that changed in 1998–99, when the results of two independent cosmological surveys of supernovas produced compelling evidence that the rate of expansion of the Universe is increasing with time and that the spatial geometry of the Universe was flat ($k = 0$), even with what had seemed to be an underdensity of matter.⁷

Though at the time this was seen as a shocking result, for years there had been mounting evidence that something was seriously amiss with standard cosmological models. Without a process to increase the rate of the Universe’s expansion with time, the large measured value of H_0 consistently gave an embarrassingly short lifetime for the Universe, significantly less than the reliably inferred ages of the oldest stars! (The current stellar record holder is HE 1523-0901, coming in at a spry 13.2 billion years.) People were aware that a cosmological constant could fix this, but to get an observationally reasonable balance between the energy density of ordinary matter and a vacuum energy density seemed like a desperate appeal to “it-just-so-happens” fine-tuning. But the new millennium brought with it unambiguous evidence that this is the way things are: our Universe is about 30% nonrelativistic matter and 70% vacuum energy, and boasts a Euclidean spatial geometry. So right now *it just so happens* that there is a bit more than twice as much energy in the vacuum as there is in

⁷Perlmutter et al. 1998 *ApJ*, 517, 565; Riess et al. 1999 *Astron J.*, 116, 1009.

ordinary matter. Nobody has the foggiest idea why; for most of the age of the Universe, one or the other will be completely dominant.

10.6.1 A Universe of ordinary matter and vacuum energy

It is possible to find a rather simple mathematical function for the scale factor $R(t)$ of our Universe. With $E = 0$ for a Euclidean space, equation (10.94) is

$$\dot{R} = \left(\frac{8\pi G \rho}{3} \right)^{1/2} R. \quad (10.108)$$

The energy density ρc^2 is a combination of nonrelativistic matter ρ_M , which decays as $1/R^3$, and a vacuum energy density ρ_V , which remains constant. With ρ_{M0} the current value of ρ_M and $R_0 = 1$, we have $\rho_M = \rho_{M0}/R^3$, and therefore

$$\rho R^2 = \rho_V R^2 + \frac{\rho_{M0}}{R}. \quad (10.109)$$

Substituting equation (10.109) into equation (10.108) leads to

$$\frac{R^{1/2} dR}{\sqrt{\rho_{M0} + \rho_V R^3}} = \sqrt{\frac{8\pi G}{3}} dt. \quad (10.110)$$

Integration then yields

$$\int \frac{R^{1/2} dR}{\sqrt{\rho_{M0} + \rho_V R^3}} = \frac{2}{3} \int \frac{d(R)^{3/2}}{\sqrt{\rho_{M0} + \rho_V R^3}} = \frac{2}{3} \frac{1}{\sqrt{\rho_V}} \operatorname{asinh} \left(\sqrt{\frac{\rho_V}{\rho_{M0}}} R^{3/2} \right) = \sqrt{\frac{8\pi G}{3}} t, \quad (10.111)$$

where asinh denotes the inverse sinh function. Equation (10.111) then tells us

$$R^{3/2} = \sqrt{\frac{\rho_{M0}}{\rho_V}} \sinh \left(\frac{3}{2} \sqrt{\frac{8\pi G \rho_V}{3}} t \right). \quad (10.112)$$

In terms of the Ω parameters, we have

$$\Omega_M = \frac{8\pi G \rho_{M0}}{3H_0^2}, \quad \Omega_V = \frac{8\pi G \rho_V}{3H_0^2}; \quad (10.113)$$

henceforth, we drop the 0 subscript on the Ω 's with the understanding that these parameters refer to current time. The dynamic equation of motion (10.108) at the present epoch tells us directly that

$$\Omega_M + \Omega_V = 1. \quad (10.114)$$

In terms of the observationally accessible quantity Ω_M , the scale factor R becomes

$$R = \frac{1}{(\Omega_M^{-1} - 1)^{1/3}} \sinh^{2/3} \left(\frac{3}{2} \sqrt{1 - \Omega_M} H_0 t \right), \quad (10.115)$$

or, with $\Omega_M = 0.31$,

$$R = 0.7659 \sinh^{2/3}(1.246 H_0 t). \quad (10.116)$$

This gives a current age of the Universe t_0 of

$$H_0 t_0 = \frac{2 \operatorname{asinh}(\sqrt{\Omega_M^{-1}} - 1)}{3\sqrt{1 - \Omega_M}} = \frac{\operatorname{asinh}[(0.7659)^{-3/2}]}{1.246} = 0.9553, \quad (10.117)$$

i.e., t_0 is rather close to $1/H_0$: $t_0 \simeq 13.8$ billion years. (This calculation ignores the brief period of the Universe's history when it was radiation dominated.)

Exercise 10.6.1

Show that the redshift z is related to the time t since the big bang by

$$z = 1.306 \sinh^{-2/3} \left(\frac{1.1903t}{t_0} \right) - 1,$$

with inverse

$$\frac{t}{t_0} = 0.84012 \operatorname{asinh} \left(\frac{1.4919}{[1+z]^{3/2}} \right) \rightarrow \frac{1.2534}{(1+z)^{3/2}},$$

with the final limit corresponding to large z . If a civilization develops 5 billion years after the big bang and we detect their signals(!), at what redshift will they be coming from?

Exercise 10.6.2

If we write R in the form of $\alpha \sinh^{2/3} \beta t$, show that the Hubble parameter is given by

$$H = \frac{2\beta}{3} \coth \beta t = \frac{2\beta}{3} \sqrt{1 + \alpha^3(1+z)^3}.$$

Does this make sense for $t = t_0$? What is the interpretation of $z = -1$ in this formalism?

Figure [10.4] is a comparison of a classical open model, with $\Omega_M = 0.31$, with our vacuum-dominated Universe for the same value of Ω_M . Notice how the extended transition from concave to convex temporal growth causes our Universe to grow nearly linearly with time over much of its history. The open model is some 12% younger than the now-accepted vacuum-dominated model and would not be old enough to accommodate the oldest known stars.

Figure [10.5] summarizes the $\Omega_V - \Omega_M$ parameter space. Accelerating/decelerating and open/closed boundaries are also shown. The pioneering supernova data have now been

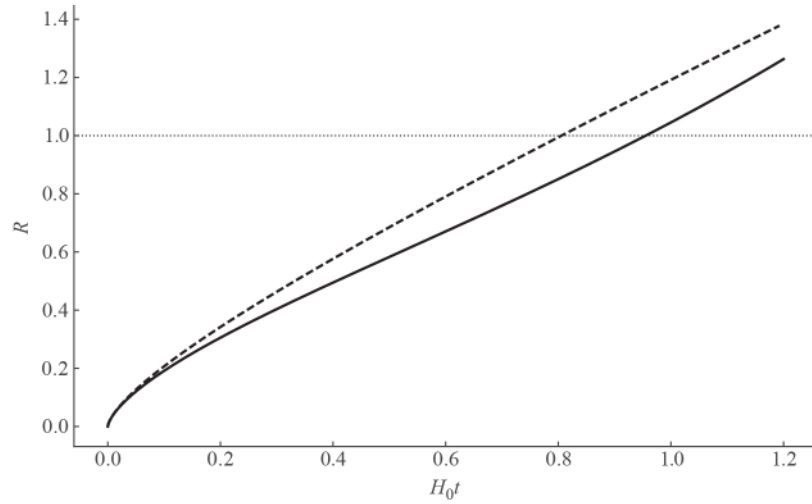


Figure 10.4 Comparison of $R(t)$ versus $H_0 t$ for our Universe eq. (10.115), (solid black curve) and an open universe eq. (10.107), (dashed curve), both with $\Omega_M = 0.31$. Crossing of the dotted line $R = 1$ (at the current age) occurs earlier for the positive energy open universe, which is about 12% younger than our Universe. This would be too young for the oldest stars.

superseded by statistical analyses of the cosmic microwave radiation background fluctuations. These more recent results both confirmed the original supernova date interpretation of an accelerating Universe and more tightly constrained the allowed parameter range. But at a more refined level, there is also a new source of tension in the data: the cosmic background radiation and supernovas seem to be giving not quite consistent values for H_0 . This is discussed further in §10.8.

Figure [10.6] is a plot of equation (10.117), showing the age of vacuum/matter universes as a function of Ω_M . The smaller Ω_M is, the older the inferred age of the universe. As $\Omega_M \rightarrow 0$, $H_0 t_0$ approaches $-(\ln \Omega_M)/3$, a phenomenon that Arthur Eddington, an early proponent of such models on philosophical grounds, dubbed the “logarithmic eternity.”

Exercise 10.6.3

The $\sinh^{2/3}(x)$ function can be awkward to work with when inside an integral. There is, however, a simple and accurate numerical approximation,

$$\sinh^{2/3}(x) \simeq \frac{x^{2/3}}{1 - x^2/9},$$

with a maximum error of about 2% over the range of x relevant for the age of the Universe, corresponding to $0 \leq x \leq 1.19$. (This maximum error occurs at $t = t_0$.) Using this approximation in our expression in equation (10.116) for $R(t)$, determine the current size of the horizon in our Universe, and compare it with the E-dS value $2c/H_0$.

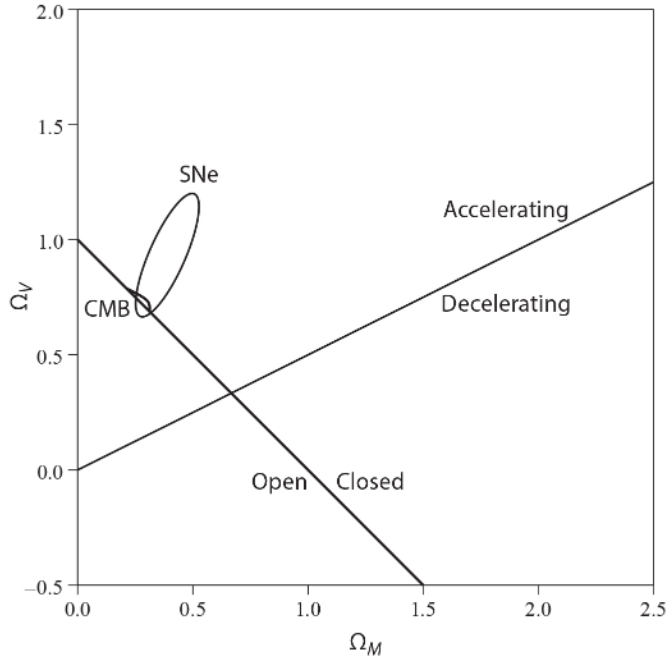


Figure 10.5 Parameter plane of Ω_V versus Ω_M assuming no radiation contribution. Regions of open/closed geometry and currently accelerating/decelerating dynamics are shown. Also shown are approximate zones of one standard deviation uncertainties for the distant supernova data (SNe) and—you have to squint—for fluctuations in the cosmic microwave background radiation (CMB), which came a decade later. Note the powerful constraint imposed by the latter. We no longer depend on the SNe data. That the Universe is accelerating is beyond reasonable doubt.

(Answer: $3.255c/H_0$; the numerically correct value is $3.259c/H_0$.) Explain qualitatively why this true horizon is larger than the E-dS value.

For a yet more accurate approximation, replace $1 - x^2/9$ with $1 - x^2/9 + 4x^4/405$, which is essentially indistinguishable from $\sinh^{2/3}(x)$ over the parameter range relevant for the lifetime of the Universe.

10.6.2 Radiation-dominated era

The early stages of the Universe were dominated by radiation, by which we mean not only photons but very relativistic particles as well. The energy density of this radiation is denoted by $\rho_\gamma c^2$. Under these conditions,

$$\rho_\gamma \propto 1/R^4.$$

This follows from the fact that the equilibrium energy density of thermal radiation is proportional to T_γ^4 , where T_γ is the radiation temperature. The temperature T_γ must have the same

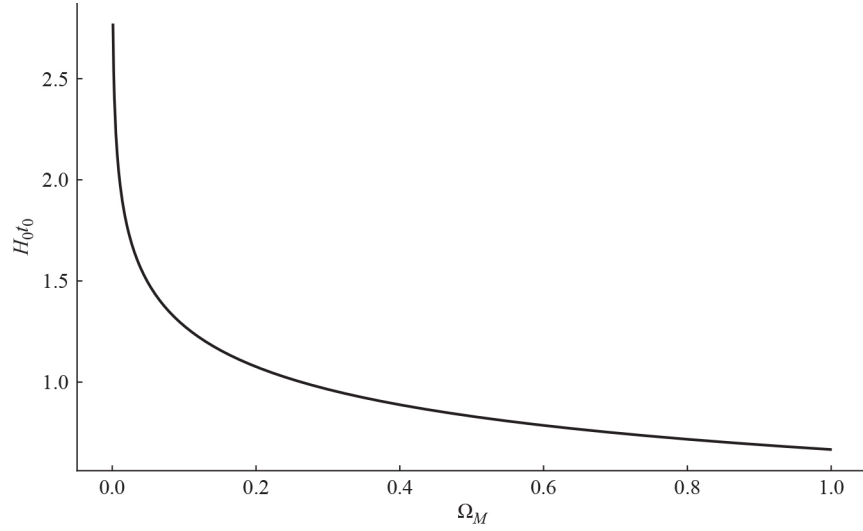


Figure 10.6 The current age of the Universe $H_0 t_0$ as a function of Ω_M for flat, matter plus vacuum energy models with negligible radiation, equation (10.117). As Ω_M approaches unity, the model recovers the Einstein–de Sitter value $H_0 t_0 = 2/3$; as $\Omega_M \rightarrow 0$, $H_0 t_0$ becomes proportional to $-\ln \Omega_M$ (Show!), and we recover the “logarithmic eternity” that was first highlighted by Sir Arthur Eddington.

R -scaling as a photon with energy $h\nu$: both the frequency ν as well as T_γ (which measures average energy per particle) decrease as $1/R$. In these early times, the dynamic equation of motion (10.94) is dominated by the two terms on the left side of the equation, both of which are very large (becoming infinite as $t \rightarrow 0$) compared with the constant on the right side. Then, using $\rho_\gamma R^4 = \text{constant}$, we may write the dynamical equation of motion at early times as

$$R^2 \dot{R}^2 = \frac{8\pi G \rho_\gamma R^4}{3} = (\text{constant}). \quad (10.118)$$

This leads to a simple power-law time dependence for the scale factor:

$$R(t) \propto t^{1/2} \quad (\text{radiation-dominated universe}). \quad (10.119)$$

By setting $\rho_\gamma c^2 R^4 = \rho_{\gamma 0} c^2$, the current energy density in radiation, and defining

$$\Omega_\gamma = \frac{8\pi G \rho_{\gamma 0}}{3H_0^2} \quad (10.120)$$

we obtain an explicit expression for the scale factor during the radiation-dominated era:

$$R^2 = 2\Omega_\gamma^{1/2} H_0 t. \quad (10.121)$$

In fact, using only $R \sim t^{1/2}$, if we now return to the dynamic equation of motion, we may solve explicitly for $\rho_\gamma c^2$:

$$\rho_\gamma c^2 = \frac{3c^2}{8\pi G} \left(\frac{\dot{R}^2}{R^2} \right) = \frac{3c^2}{32\pi G t^2}. \quad (10.122)$$

We thereby have an *exact* expression for what the total energy density in *all* relativistic particles must be, even if we don't yet know how many relativistic species there are. (In practice, we have a good idea of this number from other physics.)

This is a rather neat result. At a time of 1 s, we know that our Universe had an energy density of $4 \times 10^{25} \text{ J m}^{-3}$, and that is that. Moreover, since the total energy density is fixed, the greater the number of relativistic particle populations that were present in the early Universe (e.g., the various neutrino species), the *smaller* the temperature at a given time. During the epoch when hydrogen was being fused into helium and a few other low-atomic number nuclei, the production rate of these latter isotopes was sensitive to temperature. (By contrast, helium production is not so temperature sensitive, which makes a prediction of its abundance rather robust.) This temperature sensitivity of the light isotopes, combined with their observationally determined abundances, has been used to limit the number of different types of neutrinos that could have been present during the era of nucleosynthesis. We shall have more to say about this topic in §(10.8).

Exercise 10.6.4

Show that the equation for $R(t)$ that spans both the radiation- and mass-dominated eras is given by

$$(\dot{R})^2 = \frac{8\pi G}{3} \left(\frac{\rho_{M0}}{R} + \frac{\rho_{\gamma 0}}{R^2} \right).$$

Integrate this equation from time zero to time t to obtain

$$(R + I)^{3/2} - 3I(R + I)^{1/2} + 2I^{3/2} = \frac{3\Omega_M^{1/2} H_0 t}{2},$$

where $I = \rho_{\gamma 0}/\rho_{M0} \simeq 1.75 \times 10^{-4}$. From this equation, a reduced cubic in $(R + I)^{1/2}$, show that the correct expressions for the radiation- and mass-dominated scale factors are recovered in the small- R and large- R limits, respectively.

Exercise 10.6.5

It is possible to solve the reduced cubic polynomial of exercise 10.6.4 explicitly for $R(t)$, but the solution is cumbersome to work with. Instead we shall work directly with $H_0 t$. It is possible, in fact, to build construct an explicit and accurate formula that is valid for the almost the entire history of the Universe, from the early radiation era through the vacuum dominated future. This is done by using a matched asymptotic expansion.

- a) Equation (10.115) is an expression for R that spans both the matter and vacuum dominated eras. Show that this equation is valid provided that $\rho_{\gamma 0} \ll \rho_{M0} R$, or

$$R \gg \frac{\rho_{\gamma 0}}{\rho_{M0}} \simeq 1.75 \times 10^{-4}, \text{ or } \Omega_M^{1/2} H_0 t \gg \frac{2}{3} \left(\frac{\rho_{\gamma 0}}{\rho_{M0}} \right)^{3/2} \simeq 1.54 \times 10^{-6},$$

the era for which we may ignore radiation. Next, show that the radiation-plus-matter solution is valid when $\rho_{M0} \gg \rho_V R^3$, or

$$R \ll \left(\frac{\rho_{M0}}{\rho_V} \right)^{1/3} \simeq 0.77, \text{ or } \Omega_M^{1/2} H_0 t \ll \frac{2}{3} \left(\frac{\rho_{M0}}{\rho_V} \right)^{3/2} \simeq 0.2.$$

Hence, there is an intermediate asymptotic regime, $O(10^{-4}) \ll R \ll O(1)$ (redshifts between 10^4 and order unity), and $O(10^{-6}) \ll \Omega_M^{1/2} H_0 t \ll O(1)$ where both of our solutions are valid: the matter dominated era.

b) Consider the following expression for $1/(H_0 t)$:

$$\frac{1}{\Omega_M^{1/2} H_0 t} = \frac{3}{2} \left[\frac{1}{(R+I)^{3/2} - 3I(R+I)^{1/2} + 2I^{3/2}} - \frac{1}{R^{3/2}} + \frac{\sqrt{\Omega_M^{-1} - 1}}{\operatorname{asinh} \left(R^{3/2} \sqrt{\Omega_M^{-1} - 1} \right)} \right]$$

Note that this consists of the sum of the reciprocals of our scale factor solutions at earlier (radiation-to-matter) and later (matter-to-vacuum) eras, minus the reciprocal solution of the overlapping (matter) era. Why is it necessary to add solutions reciprocally as opposed to directly, and to subtract the overlapping form? Show that this formula will reproduce the correct leading order behavior for the $t(R)$ functional relationships at all times. A comparison of the exact numerical solution for $H_0 t$ with the above asymptotic approximation is given in Table (10.1).

Table 10.1 Comparison of $H_0 t$ as a function of R , taken from the asymptotic formula of exercise 10.6.5, with a direct numerical solution of the full differential equation [10.108] including radiation, matter, and vacuum contributions. We have used in our comparison $\Omega_M = 0.31$, $\Omega_V = 0.69$ and $\Omega_{\gamma_0} = 9.0448 \times 10^{-5}$ based on a current photon temperature of 2.7255 K, with an additional neutrino contribution of 68.13% (see §10.9.3). The uniform accuracy of the asymptotic solution is striking.

Scale Factor R	$H_0 t$ (Asymptotic Eq.)	$H_0 t$ (Exact Integral)	Rel. Error
1.00E-06	5.251406E-11	5.251405E-11	1.10E-07
1.00E-05	5.198469E-09	5.198468E-09	1.08E-07
1.00E-04	4.751339E-07	4.751339E-07	9.06E-08
1.00E-03	2.985726E-05	2.985726E-05	3.84E-08
1.00E-02	1.155763E-03	1.155763E-03	1.27E-08
1.00E-01	3.769606E-02	3.769603E-02	9.42E-07
3.00E-01	1.945586E-01	1.945568E-01	9.22E-06
1.00E+00	9.549512E-01	9.548746E-01	8.02E-05

10.7 Observational foundations of cosmology

10.7.1 The discovery of cosmological redshifts

Vesto Melvin Slipher is not a name likely to be familiar to readers not already well versed in the history of cosmology, but in the early twentieth century, he was one of the most skilled observational astronomers in the world. If we allow that Edwin Hubble was the father of modern observational cosmology, then Slipher surely deserves the title of grandfather.

In 1912, using a 24-inch reflecting telescope, Slipher was the first person to measure the redshifts of external galaxies. He didn't know that was what he had done, because the notion of galaxies external to our own was not one that was well formed at the time. This was, nevertheless, a remarkable achievement. Nebular spectroscopy was very hard, tedious work, spreading out and thereby diminishing the light from the already very faint, low-surface-brightness smudges of spiral nebulae through highly dispersive glass prisms. Slipher worked at Lowell Observatory, a small and rather isolated private outpost in Flagstaff, Arizona. Percival Lowell, Slipher's boss, was at the time completely obsessed with mapping what he thought were the Martian canals!

Slipher carried out his lonely and unenviable task in a state of scientific isolation. But by 1922, he had accumulated 41 spectra of spiral nebulae, of which 36 showed a shift toward the red end of the spectrum. He had no way of organizing these data to reveal the linear scaling of the redshifts with distance, since he had no idea what the distances to the nebulae were. But other observations were at this time showing an apparent trend of greater redshift with fainter nebulae. In the end, the decisive step was taken not by Slipher, but by Edwin Hubble.

With the aid of his able assistant Milton Humason, Hubble firmly established a *linear* relationship between nebular distance and redshift.⁸ How did Hubble manage this amazing feat? Using the new and extraordinary 100-inch telescope on Mount Wilson, Hubble had earlier resolved individual Cepheid variable stars, whose distances *could* be measured, in the outer arms of the Andromeda spiral *galaxy*, as we shall henceforth refer to it. In obtaining the distances to the nebulae, Hubble also showed that these photographic smudges were, in fact, true galaxies in their own right.

At the time, Cepheid variable stars were very well studied in our own Milky Way Galaxy. Cepheids oscillate in brightness with regular periods ranging from days to months. What makes them an important candle for astronomers is that they have a well-defined relationship between their oscillation time period and their absolute mean luminosity. So, to determine an absolute distance, you first find a distant Cepheid variable, which is a rather bright star, and measure its oscillation period and determine thereby its true luminosity. Next, measure the star's flux, which is simply the energy per unit area per unit time crossing your detector. The flux is the absolute luminosity L divided by $4\pi r^2$, where r is the distance to the star. Therefore, by directly measuring the flux, and inferring L from the oscillation period, you may deduce r , the distance to the star. This is all very simple in principle, but for the kinds of distances we are discussing here, you would need a superb-quality, huge (100-inch) telescope. To such a state-of-the-art instrument, only Hubble and a very small handful of other astronomers had access at the Mount Wilson Observatory in Southern California.⁹

⁸Hubble, E. 1929, *PNAS*, 15,168.

⁹Much later, in 1952, Hubble's quantitative results were found to be inaccurate. It turns out that Cepheids come in two quite separate populations, with very different period-luminosity relations! This was discovered by W. Baade,

10.7.2 The cosmic distance ladder

As observations improved through the 1930s, the linear relation between velocity v and distance r , which subsequently became known as the *Hubble law*, $v = H_0 r$, continued to be more firmly established. There are two principal difficulties with collecting data in support of the Hubble expansion.

First, actual galaxies need not be moving perfectly with the Hubble expansion (or “Hubble flow”): their motions will be affected by neighboring masses. The best-known example of this is the Andromeda galaxy, whose redshift is, in fact, a blueshift! It is approaching our own Milky Way Galaxy at about 300 km s^{-1} . The redshift measurement problem is greatest for a nearby galaxy whose “peculiar velocity” (deviation from Hubble flow) could easily be a large fraction of, or indeed dominate, its Hubble recession velocity. Second, it is very difficult to establish distances to cosmological objects. We can establish distances to nearby objects relatively easily, but these galaxies are also precisely those that are affected by large peculiar velocities. The galaxies unaffected by large peculiar velocities are just the ones whose (much greater) distances are difficult to establish!

But observational astronomers are resourceful, and they have come up with a number of clever techniques which have served them well. The idea is to create a cosmic “distance ladder,” which is perhaps better described as a linked chain. They start with direct established measurements of certain carefully chosen objects and then use those findings to calibrate the distances to other, more distant, objects. Then, they repeat the process. The following describes how it actually works.

We start with our solar system. Today, we can bounce radar signals off planets and measure the time of flight (even testing general relativity in the process, as we have seen) to get extremely accurate distances. Next, we make use of our precise knowledge of the astronomical unit (denoted by AU) thus obtained to use the classic technique of trigonometric parallax. Earth’s motion around the Sun creates a baseline of 2 AU, which allows us to have a different perspective on nearby stars as we orbit. We see nearby stars shift in angular position on the sky relative to their much more distant counterparts. The observed angular shift is inversely proportional to the distance of the nearer star from the solar system. We define the unit of distance known as a parsec (pc) as the distance at which a separation of 1 AU subtends 1 second of arc ($1''$). We can turn this definition around and say that the parallax angle, perversely denoted by π in the astronomical literature and measured in arc seconds, corresponds to a distance to a measured star of $1/\pi$ pc.

The next rung up the ladder is known as spectroscopic parallax. There is no actual parallax angle in this case, it is just an analogous name. The idea is that we first use the method of direct trigonometric parallax to obtain distances to stars with well-understood radiation spectra, that it is to say stars of a *given spectral type*. Knowing the distance to a nearby star of a given spectral type and measuring its flux, we may deduce its intrinsic luminosity L , since $F = L/4\pi r^2$ where F is the measured flux and r the measured distance to the star. Suppose now that we see a star with precisely the same spectrum but much fainter and therefore more distant. We cannot measure its parallax this time; it is too far away. But the detailed agreement between spectra means this is the same type of star we’ve already measured, with

who was then motivated to introduce the concept of distinct stellar populations (distinguished by ages) into astronomy. This revised the entire extragalactic distance scale but fortunately kept intact the linearity of the redshift-distance relation.

the same mass and same luminosity. We then measure the new star's flux F , and deduce *its* distance.

We keep going. Distances to Cepheid variables, which are very bright, can be calibrated by spectroscopic parallax, and these stars have, as explained above, a well-defined period-luminosity relationship. Cepheids are bright enough that they can be seen individually in external galaxies with distances up to 50 Mpc. We measure, thereby, the distances to these external galaxies.

We keep going. In studying the properties of external galaxies, Tully and Fisher showed that the rotation velocities of spiral galaxies (measured by the Doppler shifts in the combined spectra of moving stars) were tightly correlated with the intrinsic luminosity of the galaxy.¹⁰ This is not terribly surprising in itself: the larger the stellar mass of a galaxy, the larger the luminosity; and the larger the binding mass, the larger the rotational velocity. Tully and Fisher crafted this notion into a widely used tool for establishing the distances to very distant galaxies. Elliptical galaxies, which are supported by the dispersion of stellar velocities rather than their systematic rotation, were also turned into useful distance indicators. Here the correlation between luminosity and velocity *dispersion* is known as the Faber-Jackson relation.¹¹

The final step in the cosmic distance ladder involves Type Ia supernovas. Type Ia supernovas are thought to occur when a white dwarf in a tight binary system accretes just enough matter from its companion to tip itself over the Chandrasekhar mass. (The reader will recall that this is the maximum possible mass a white dwarf can sustain by electron degeneracy pressure, about 1.4 times the mass of the Sun. See §8.4.) When this mass is exceeded, the white dwarf implodes, overwhelmed by its now unsupportable self-gravitational attraction. In the process, carbon and oxygen nuclei are converted to ^{56}Ni , triggering a thermonuclear explosion that can be seen quite literally across the Universe. The beauty of type Ia supernovas, from an astronomical perspective, is that they always occur in a white dwarf of nearly the same mass. Therefore, there is little variation in the absolute intrinsic luminosity of the supernova explosion. To the extent that there is some variation, it is reflected not just in the luminosity but in the rise and decay times of the emission, the “light curve.” The slower the decline, the larger the luminosity. Astronomers can correct for this. This relation has therefore been well calibrated in many galaxies with well determined distances.

The Type Ia supernova data were the first to provide compelling evidence that the Universe is accelerating. To understand how this was obtained we need to return to our formula for the flux, $F = L/4\pi r^2$, and understand how this must be modified in an expanding, possibly even curved, spatial geometry.

10.7.3 The parameter q_0

Historically, observational astronomers generally worked with distances for which the time it has taken light to reach us from the source is small compared with the age of the Universe. The idea was to measure small, leading-order changes from conditions at the current epoch, $t = t_0$. Under these circumstances, it is mathematically convenient to expand our basic cosmological functions $R(t)$ and $H(t)$ in a Taylor series in $t - t_0$. For example, we may

¹⁰Tully, R. B., & Fisher, J. R. 1977, *Astron. Astrophys.*, 54, 3, 661.

¹¹Faber, S. M., & Jackson, R. E. 1976, *ApJ*, 204, 668.

expand the Hubble parameter $\dot{R}/R$ in time:

$$H(t) = \frac{\dot{R}}{R} = H_0 + (t - t_0) \left(\frac{\ddot{R}_0}{R_0} - H_0^2 \right) + \dots, \quad (10.123)$$

since $\dot{R}^2/R^2$ at the present time t_0 is H_0^2 . (Recall that $t < t_0$ for observed sources.) We may also write this expression as

$$H(t) = H_0 + H_0^2(t_0 - t)(1 + q_0), \quad (10.124)$$

where we have defined the *deceleration parameter* q_0 by

$$q_0 \equiv -\frac{\ddot{R}_0 R_0}{\dot{R}_0^2} \rightarrow -\frac{\ddot{R}_0}{H_0^2} \quad \text{for } R_0 = 1. \quad (10.125)$$

This is an important observational quantity because it embodies the observed evolution of the Hubble parameter. Next, we use $(1+z)R(t) \equiv 1$ and expand $R(t)$ in a similar Taylor series around t_0 . Then, to leading order in z , we obtain $z = H_0(t_0 - t)$, and then equation (10.124) becomes

$$H(z) = H_0[1 + z(1 + q_0)]. \quad (10.126)$$

For the classical, decelerating matter-dominated cosmological models, q_0 is defined in such a way that it should be a positive quantity. If $q_0 < 0$, the Universe would be accelerating, a seemingly outlandish possibility not taken very seriously until after 1998. Consider the integral defining the comoving coordinate of a photon, $r = \int c \, dt/R(t)$. With $R_0 = 1$ and $t - t_0 \equiv \delta t$, we carry out the Taylor expansion in δt of

$$1 + z = \frac{1}{R(t)} = \frac{1}{R(t_0 + \delta t)} = 1 - \delta t \dot{R}_0 + (\delta t)^2 \left(\dot{R}_0^2 - \frac{\ddot{R}_0}{2} \right) + \dots \quad (10.127)$$

To linear order this gives $\delta t = -z/\dot{R}_0 = -z/H_0$. Using this result in the small quadratic correction term in $(\delta t)^2$ in equation (10.127), the next-order refinement is

$$-\delta t = t_0 - t = \frac{1}{H_0} \left[z - \left(1 + \frac{q_0}{2} \right) z^2 \right]. \quad (10.128)$$

Using our expansion for $1/R$ we find

$$r(t) \equiv c \int_t^{t_0} \frac{dt'}{R} = c(t_0 - t) + \frac{cH_0}{2}(t - t_0)^2 + \dots, \quad (10.129)$$

and substituting with the help of equation (10.128) through order z^2 :

$$r(z) = \frac{c}{H_0} \left[z - \frac{z^2}{2} (1 + q_0) + \dots \right]. \quad (10.130)$$

Equation (10.130) is general for any FRW model. The parameter q_0 embodies the leading-order deviations from a simple Euclidean model in which $r(z) \propto z$.

10.7.4 The redshift–magnitude relation

For large redshifts z , the flux that is measured from a source at cosmological distances differs significantly from its Euclidean form $L/4\pi r^2$, for several important reasons. It is best simply to write the answer and then explain the appearance of each modification. We keep factors of R_0 present, so that general FRW metrics are permitted.

The flux from an object at redshift z is given by

$$\mathcal{F}(z) = \frac{LR^2(t)}{4\pi R_0^4 r^2(z)} = \frac{L}{4\pi(1+z)^2 R_0^2 r^2(z)}, \quad (10.131)$$

where $r(z)$ is computed, as always, by $c \int_t^{t_0} dt'/R(t')$, with this definite integral converted to a function of z for the cosmological model at hand. (Recall $R(t)/R_0 = 1/(1+z)$, relating t to z .) The luminosity distance is then defined by $\mathcal{F}(z) \equiv L/4\pi d_L^2$ or

$$d_L(z) = (1+z)R_0 r(z), \quad (10.132)$$

which reduces to the conventional Euclidean distance at small z .

Explanation: the two factors of $1+z$ in the denominator of equation (10.131) arise from the change in the luminosity L . First, the photons are emitted with redshifted energies. But even if we were measuring only the rate at which the photons were being emitted (like bullets), we would require an additional second $1+z$ factor, quite separate from the first, due to the emission-interval time dilation. The photons thus have less energy and are emitted less often. The proper radius of the sphere over which the photons from the distant source at z are now distributed is $R_0 l(z)$, where R_0 is, as usual, the *current* value of $R(t)$. For the Einstein–de Sitter universe with $R_0 = 1$, the explicit formula is

$$\mathcal{F}(z) = \frac{LH_0^2}{16\pi c^2(1+z)(\sqrt{1+z}-1)^2} \quad (\text{Einstein–de Sitter}). \quad (10.133)$$

The simple Euclidean value of $L/4\pi r^2$ is recovered at small z by recalling $cz = v = H_0 r$, whereas for large z , all photons come from the current horizon distance $2c/H_0$, and the $1/z^2$ behavior in $\mathcal{F}$ is due entirely to $1+z$ cosmological redshifts.

Exercise 10.7.1

Show that for our Universe,

$$d_L(z) = \sqrt{\frac{3c^2}{8\pi G\rho_{M0}}} (1+z) \int_0^z \frac{dx}{\sqrt{\rho_V/\rho_{M0} + (1+x)^3}},$$

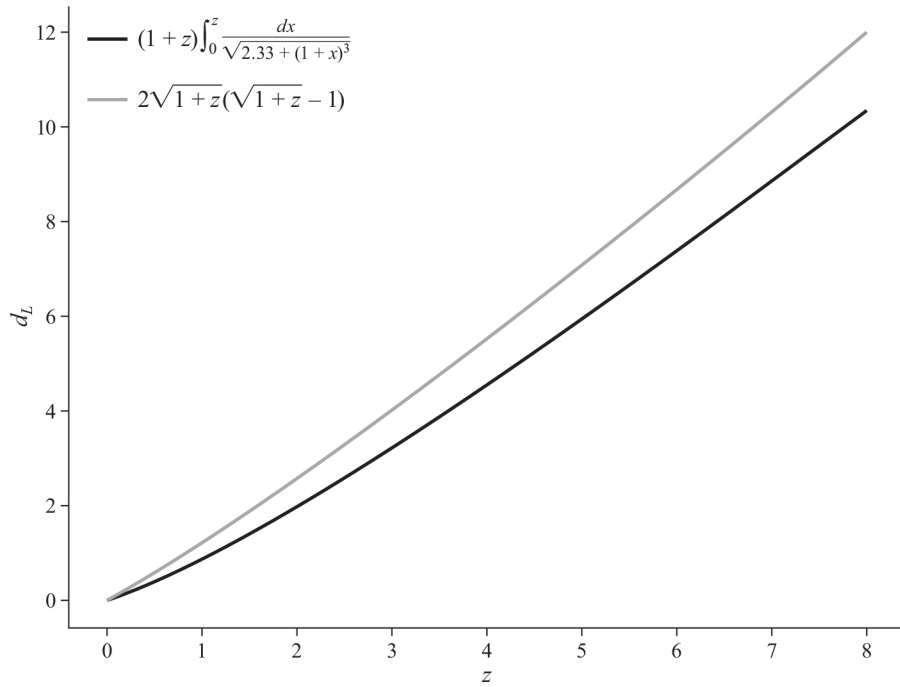


Figure 10.7 A comparison of the luminosity distance d_L as a function of redshift z for an Einstein–de Sitter universe (gray) and our Universe (black), for $\rho_V/\rho_{M0} = 2.33$. The units of d_L are $\sqrt{3c^2/8\pi G\rho_{M0}}$.

where ρ_V/ρ_{M0} is the ratio of the vacuum to current mass energy densities. In figure (10.7) we show a comparison of $d_L(z)$ (in units of $\sqrt{3c^2/8\pi G\rho_{M0}}$) for an Einstein–de Sitter universe and our own with an energy density ratio of 2.33.

A “magnitude” is an astronomical conventional unit, used for rather arcane historical reasons. It is a logarithmic measure of the flux. Explicitly,

$$\mathcal{F} = \mathcal{F}_0 10^{-0.4m}, \quad (10.134)$$

where $\mathcal{F}$ is the measured flux and $\mathcal{F}_0$ is a constant that depends upon which wavelength range is being measured. This defines m as the apparent magnitude. (Note: A larger magnitude is *fainter*!) The “bolometric magnitude” covers a wide wavelength range and is considered a measure of the total flux; in that case $\mathcal{F}_0 = 2.52 \times 10^{-8} \text{ J m}^{-2}$. The idea behind a redshift-magnitude test is to plot m versus z for many objects that (ideally) have the same intrinsic luminosity, like type Ia supernovae. The observer then deduces which $d_L(z)$ emerges when the results are fit to equation (10.131) for an FRW model. It was this process that led to the discovery in 1998–99 by Perlmutter, Riess, and Schmidt that our Universe must have a large effective value of the vacuum energy density, as in exercise 10.7.1: we live in an accelerating Universe. (See figure [10.8].)

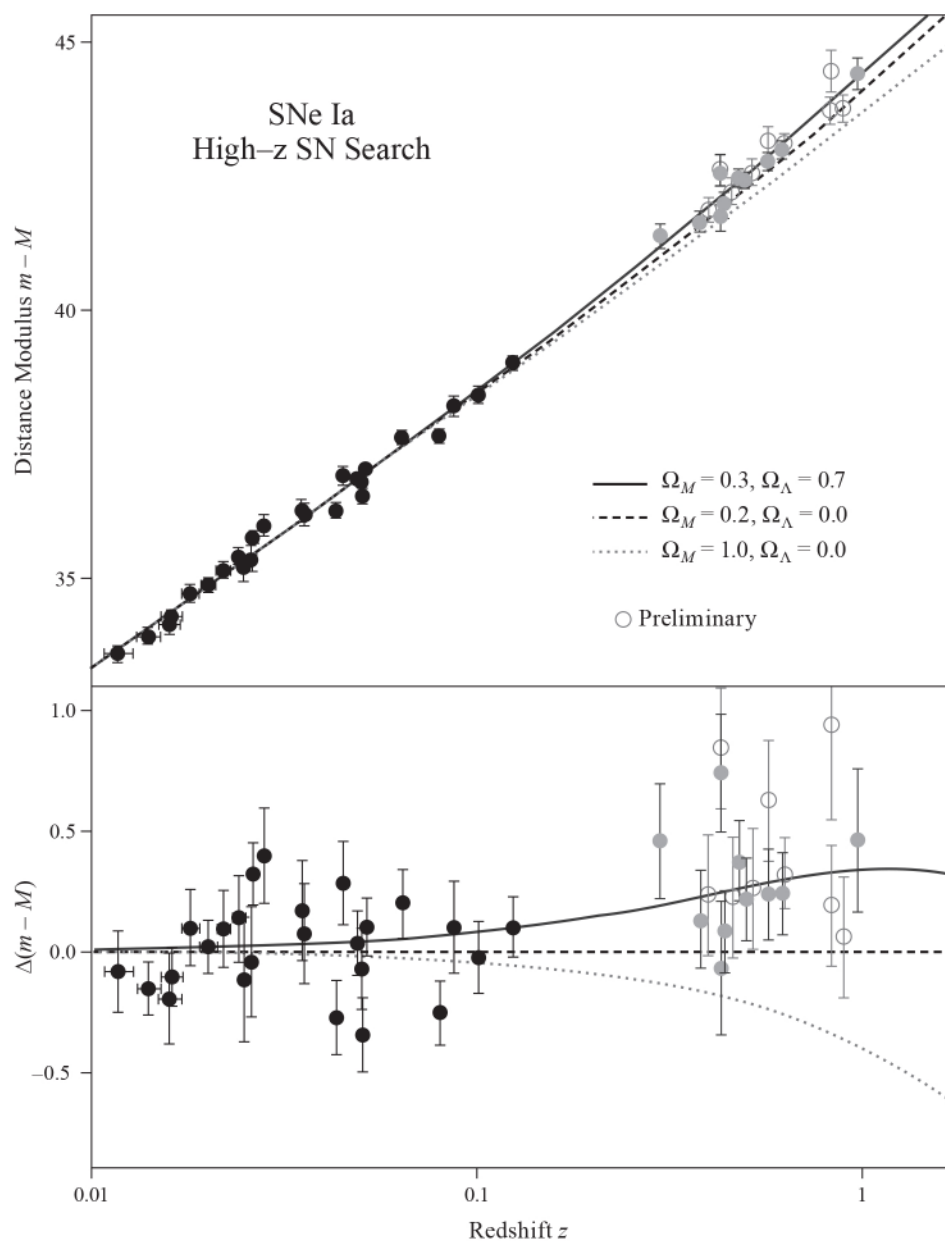


Figure 10.8 Evidence for an accelerating Universe from type Ia supernovae. *Top:* A redshift-magnitude plot for three different FRW models, $\Omega_M = 1, \Omega_V = 0$, an Einstein-de Sitter model (dotted line); $\Omega_M = 0.2, \Omega_V = 0$, an open model (dashed line); and $\Omega_M = 0.3, \Omega_V = 0.7$, an accelerating model (solid line). *Bottom:* The same plot with the inverse square slope removed. The data are much better fit by the accelerating model.

Exercise 10.7.2

Show that

$$\mathcal{F}(z) = \frac{LH_0^2}{4\pi c^2 z^2} [1 + (q_0 - 1)z + \dots]$$

for any FRW model. Thus, with knowledge of L , observers can read off the value of H_0 from the dominant $1/z^2$ leading-order behavior of the redshift-magnitude data, but knowledge of q_0 comes only once this leading-order behavior has been subtracted from the data.

With the determination of Ω_V , the classical problem of the large-scale structure of the Universe has been solved. Six quantities were to be determined:

- The Hubble constant, $H_0 = \dot{R}/R$ at the present epoch.
- The age of the Universe, t_0 .
- The curvature of the Universe, in essence the integration constant $2E = -c^2/a$.
- The ratio Ω_M of the current mass density ρ_{M0} to the critical mass density $3H_0^2/8\pi G$.
- The value of the cosmological constant or, equivalently, the current ratio Ω_V of the vacuum energy density $\rho_V c^2$ to the critical energy density.
- The value of the q_0 parameter.

Within the context of FRW models, these parameters are not completely independent but are related by the dynamic equations for $\dot{R}$ (10.82) and $\dot{R}^2$ (10.94). A quick summary:

- $H_0 \simeq 67.6 \text{ km s}^{-1} \text{ Mpc}^{-1}$
- $t_0 \simeq 13.8 \text{ billion years}$
- $E \simeq 0$
- $\Omega_M \simeq 0.31$
- $\Omega_V \simeq 0.69$.
- $q_0 = \Omega_M/2 - \Omega_V \simeq -0.54$

Exercise 10.7.3

Derive the expression given for q_0 , the final result in this list. This means that the expression in square brackets in exercise 10.7.2 for $\mathcal{F}(z)$ differs between $1 - 0.5z$ for an Einstein–de Sitter universe and $1 - 1.54z$ for our Universe, a significant difference for $z \gtrsim 0.1$.

Exercise 10.7.4

Show that for any *flat* FRW universe, if n_0 is the current number density of luminosity sources made from ordinary nonrelativistic matter, and L is the assumed constant luminosity of each source, then the total energy density in radiation is given by

$$\int_0^{t_0} L n_0 R(t) dt.$$

Show that for an Einstein–de Sitter universe this is $(2/5)n_0 L/H_0$. (There is no Olber’s paradox!) Using equations (10.116) and (10.117), show that for our Universe, under the stated assumptions, the same energy density is $0.529n_0 L/H_0$. You may use whatever sources you like to evaluate numerically the integral in question, or try using the approximation of exercise 10.6.3.

This brief list of values hardly does justice to the century-long effort to describe our Universe with precision. Because astronomers were compelled to use galaxies as “standard candles” (the astronomical term for calibrated luminosity sources), the observational measuring tools were plagued with uncertainties that could never be fully eliminated. It was the combination of establishing a reliable standard candle via the type Ia supernovae, together with the modern technical capabilities of high receiver sensitivity and automated search techniques that allowed the transitional program (The Supernova Cosmology Project) to succeed.

Since the 1998/89 breakthrough, cosmologists have not been idle. The development of extremely sensitive receivers and sophisticated modeling techniques have turned the remnant microwave radiation “noise” from the big bang itself into a vast trove of information. In particular, the statistical properties of the tiny fluctuations present in the radiation intensity—more specifically, the radiation temperature—also allow one to set very tight constraints on the the large-scale parameters of our Universe. At first, these CMB measurement results (early-time observations) extracted from the missions known as WMAP and Planck, seemed consistent with the supernova data (much later epochs). (See figure [10.5].) More recently, a significant tension has arisen between these two data sets over the value of the Hubble constant H_0 . Whether this is a matter of tracking down sources of error, or an indication of new physics, is not clear at present. The latest data from the James Webb Space Telescope, however, appear to point toward the former, though as yet there is no clear consensus. This is discussed further in §10.8.3.

10.8 The cosmic microwave background radiation (CMB)

10.8.1 Overview

The Universe is expanding, and expanding systems cool adiabatically. That means if we follow our history backward, the Universe must have been very hot and dense. We have seen that the energy density of radiation scales as $1/R^4$, whereas the energy density of matter

(dominated by its rest mass) scales as $1/R^3$. The Universe must therefore, at some point in the past, have become radiation dominated. We shall see this occurred at a redshift $z \sim 5600$. At redshifts larger than this, not only was radiation dominant, the photons were also “well coupled” to ordinary matter. This means that a typical photon would collide many times with electrons during the interval it took the Universe to approximately double in size. These collisions kept the radiation in a state of complete thermodynamic equilibrium, so the radiation temperature did not differ from the matter temperature. In thermodynamic equilibrium, the number density of photons is *fixed*, given by the Planck function associated with a blackbody spectrum:

$$n(\nu, T) d\nu = \frac{(8\pi\nu^2/c^3) d\nu}{\exp(h\nu/k_B T_\gamma) - 1}. \quad (10.135)$$

Thus, if radiation is in thermal equilibrium with matter, there is no choice of how many photons there are at a given frequency. Mother Nature determines that precisely. It is worth dwelling on this point. Unlike a gas bag of atoms in thermal equilibrium, whose density can be whatever we like, the number density of a gas bag of photons is completely determined by its temperature, because photons can be freely created and destroyed. A spectrum may follow directly a Planck function in shape, but if the total number density is not given *exactly* by equation (10.135), the photons are simply not in thermal equilibrium with their surroundings, and their numbers will change as this thermal equilibrium is inevitably approached. So, does the unavoidable decline in photon number density with ongoing universal expansion mean that we lose a thermal equilibrium distribution for the radiation?

At about the time that the radiation energy density was becoming subdominant relative to matter, the photons also lost good coupling with the matter. Thermal equilibrium could then no longer be maintained via photon-electron collisions or induced exchanges with any atoms present. Imagine a situation in which the photons propagate with the expansion of the Universe but are otherwise unaffected by interactions. You might guess that while the overall *shape* of the frequency spectrum would be maintained, the photon number density would be diluted below that of a true blackbody. (Rather like sunlight reaching the Earth.) Not so. In fact, the number density and effective temperature of the photons together vary in just the right way to maintain an exact blackbody distribution! The ever-decreasing number density is always exactly the correct one for a true thermal Planck spectrum for the concurrently evolving and ever-cooling photon temperature.

To show this, we start with the spectrum, equation (10.135), at the time t of last scattering, by which we mean the last moment thermal equilibrium was maintained by collisions. We follow a group of photons with frequency ν and small dispersion $d\nu$ as these photons evolve with the Universe. Now, it is some much later time t' . The t' number density $n(\nu', t')$ is diluted by an overall (volume) factor of $R^3(t)/R^3(t')$ relative to the original blackbody number density at the time of last scattering t . In addition, each photon at t' that we now observe at frequency ν' must have come from a frequency way back at time t , given by $\nu = R'\nu'/R$ (where $R' \equiv R(t')$), in the original blackbody distribution. Therefore, to determine the later time t' spectrum, we follow this rule: take the original time- t blackbody distribution, dilute it by R^3/R'^3 , and then replace ν everywhere in the Planck formula by $R'\nu'/R$. Carrying this process through, we obtain

$$n(\nu', t') d\nu' = \frac{R^3}{R'^3} n(\nu, t) d\nu = \frac{R^3}{R'^3} \frac{(8\pi/c^3) (R'\nu'/R)^2 d(R'\nu'/R)}{\exp(hR'\nu'/Rk_B T_\gamma) - 1} = \frac{8\pi\nu'^2 d\nu'/c^3}{\exp(h\nu'/k_B T'_\gamma) - 1}, \quad (10.136)$$

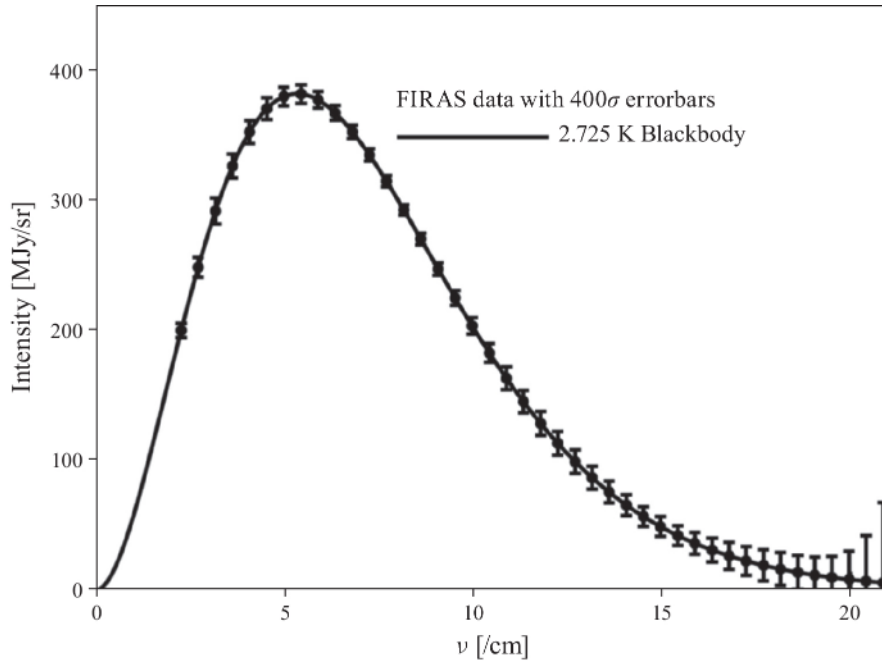


Figure 10.9 COBE satellite data showing a perfect fit to a blackbody at 2.725 K. To make them visible, the error bars are shown at 400 times their actual value! When shown at an American Astronomical Society Meeting, this plot triggered a spontaneous standing ovation. Units of ν are cm^{-1} , i.e., the wavelength in centimeters is the reciprocal of the number on the x -axis.

where $T'_\gamma = T_\gamma R/R'$, a much *cooler* temperature than T_γ . The remarkable point is that the distribution of equation (10.136) is *still* a blackbody at time t' , but with a new temperature T'_γ that has cooled in proportion to $1/R$ with the expansion. The reason this is remarkable is that there is nothing to maintain this thermal equilibrium! In fact, this scaling result follows mathematically for any number density spectrum of the form $\nu^2 F(\nu/T) d\nu$, where F is an *arbitrary* function.

Might we just be playing a mathematical game? What if, in the real Universe, other interactions scatter photons and change their frequencies in the process? For many years, there were arguments as to whether the actual cosmic radiation spectrum was truly a precise blackbody. Adding to the confusion, there were puzzling observations, largely based on strikingly nonblackbody spectra obtained from rocket-borne instruments. Now, to significantly change the radiation spectrum of the *entire* Universe would take a vast amount of energy. Whence would that have come? Theorists tied themselves into knots trying to come up with explanations. All doubts were finally erased in 1992, when the COBE (COsmic Background Explorer) satellite was launched. This satellite had submillimeter detectors of unprecedented accuracy, no rocket exhaust, of course, and it was established beyond any question that the background radiation was almost a perfect blackbody at $T_\gamma = 2.725$ K. If we assume that $\Omega_{M0} = 0.3$, this value for T_γ leads to equal energy densities of matter and radiation at $z = 5600$, as was noted earlier in this section. Figure [10.9] summarizes the data. The error bars are shown at 400 times their actual value, just to make them visible! (The earlier rocket observations had by this time been retracted.)

Notice the careful phrasing: “almost a perfect blackbody.” The photon distribution is, in fact, not *exactly* a blackbody for two very interesting reasons. The first is that Earth is not at rest relative to the CMB frame. In fact, the local group of galaxies appears to be moving at about 630 km s^{-1} relative to the CMB, a surprisingly large value. This translates to a measured radiation temperature that is about $2 \times 10^{-3} \text{ K}$ warmer in one direction, and the same amount cooler in the opposite direction—in effect a temperature dipole. This induces no change in the overall energy of the radiation.

The second reason is much deeper in its physics. The objects that our Universe comprises collapsed out of an expanding gaseous background, forming great clusters of galaxies and, ultimately, individual galaxies and stars. The seeds to form these structures could not have appeared recently: there simply hasn’t been enough time for them to have grown from tiny seeds, collapse, and then form the objects we see throughout the Universe today. The seeds must already have been present during the era when the Universe was radiation dominated, the same era when matter and radiation were still interacting strongly with one another, mainly via photon-electron scattering. This interaction would have left its imprint, at the time of last scattering (the instant before hydrogen recombined), in the form of temperature *fluctuations* in the microwave background radiation. When matter and radiation are well coupled, a compression in the matter also means a compression in the radiation, and therefore heating as well. Some regions will be a little hotter, others a bit cooler (where there has been matter expansion)—signatures of primordial acoustic waves. The simplest calculations suggested that the relative temperature fluctuation $\Delta T/T$ should have been at least $\sim 10^{-3}$, for these initially tiny compressions to be able to form nonlinear structures by the current epoch.

COBE indeed found the temperature fluctuations (figure [10.10]). But when COBE (and later satellites in yet greater detail) did find them, they were an order of magnitude smaller than expected. So how does structure form in the Universe? The consensus is that there is considerably more matter in the Universe than the usual atomic nuclei and electrons we know about. The Universe is also pervaded by what is known as dark matter, which we encountered earlier in this chapter, and it turns out to be most of the particulate matter. It leaves a gravitational signature on the microwave background, but no acoustic imprint. Dark matter feels the force of gravity, and so far as is known, nothing else. Its existence had been postulated long before COBE as something that was needed to hold galaxies together. (Were visible light truly representative of the bulk of the matter, stellar velocities within galaxies would be far too large to allow the stars to be gravitationally bound.) The COBE findings offered yet another compelling line of evidence for the existence of dark matter.

The directors of the COBE project, John Mather and George Smoot, won the 2006 Nobel Prize in Physics for their work.

Exercise 10.8.1

Show that the number density of CMB photons (or of any blackbody spectrum) as a function of temperature is given by

$$\pi \left(\frac{2k_B T}{hc} \right)^3 \int_0^\infty \frac{x^2}{e^x - 1} dx = 2\pi \zeta(3) \left(\frac{2k_B T}{hc} \right)^3,$$

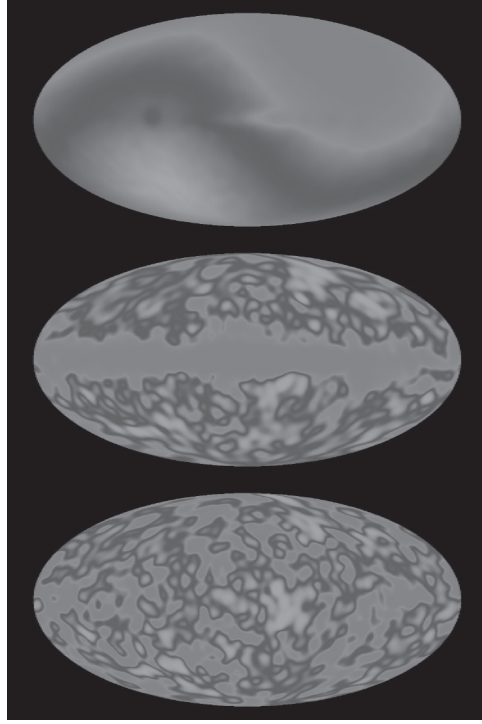


Figure 10.10 COBE temperature fluctuations. *Top*: Temperature dipole variation dominated by motion of the galaxy. *Middle*: Residual with kinematic dipole subtracted. *Bottom*: Further residual with galactic foreground subtracted. These represent primordial fluctuations in the CMB.

where $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$ is the Riemann zeta function. With $\zeta(3) = 1.202$ and a current radiation temperature of $T = 2.725$ K, show that this is about 410 photons per cubic centimeter.

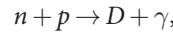
10.8.2 An observable cosmic radiation background: The Gamow argument

The idea that there was a residual radiation field after the formation of the Universe *and this radiation is potentially observable*, seems to have originated with George Gamow in the 1940s. Gamow was a theoretical physicist with a penetrating common sense instinct that allowed him to make contributions to a wide variety of important problems, from the theory of radioactivity to how DNA coding works. He spent many years developing the Big Bang theory for the origin of the Universe.¹²

¹²The name was coined by Fred Hoyle, a rival cosmologist who meant the description to be a disparaging moniker. But proponents of the theory loved the name and adopted it as their own.

Gamow drew attention to the fact that helium, with about 25% of the baryonic mass of the observable Universe, is neither an overwhelming nor a tiny constituent. Gamow was convinced, we now know correctly, that almost all this helium was made during a brief interval of nucleosynthesis in the very early Universe. (Gamow thought all the other elements were made this way, as well, but that turns out to be false. Heavier elements, along with a few percent of the helium, are made over the course of stellar evolution, with contributions from supernovae and possibly from neutron star mergers as well.) The 25% number implies that at the time of nucleosynthesis, the expansion rate of the Universe and the nuclear reaction rates could not have been very different. With a more rapid expansion rate, too much helium would be made. With a slower expansion rate, too many free electrons would decay before being packaged into helium nuclei. The delicate balance turns out to be a remarkably tight constraint, leading to a prediction that there should today be an observable residual radiation field of ~ 10 K. Let's see how it works.

At some point early in its history, the temperature of the expanding Universe evolved downward through the range of $T \sim 10^9$ K. When that happened, neutrons n and protons p were able to combine to form deuterium (a proton p plus a neutron n) nuclei. The reaction is simple:



where D is the deuterium nucleus, and γ denotes a gamma ray. With deuterium nuclei present, helium synthesis followed rapidly thereafter. Gamow assumed equal numbers of protons, neutrons and electrons at the start. As we have emphasized, to get a 25% yield of helium, the reaction rate and the age of the Universe (which is roughly the expansion time required for the Universe to double in size) should be comparable. Denote the cross section for deuterium formation as σ (units of area). Then, the reaction rate per proton due to an incoming flux $n\nu$ of neutrons (density n , velocity ν) is $n\nu\sigma$ (units of pure number per time). Roughly speaking, at the time of nucleosynthesis t , approximately one reaction per proton should have occurred. This key point is inferred from the order unity mass fraction of helium that we now observe. The heart of the Gamow argument is thus:

$$n\nu\sigma t \sim 1, \quad (10.137)$$

where t is the age of the Universe at the time of nucleosynthesis. How simple!

The product $\sigma\nu$ is just a laboratory number nearly independent of ν (because the cross section σ depends inversely on ν), and, as Gamow knew from contemporaneous nuclear physics experiments, is equal to about $4.6 \times 10^{-26} \text{ m}^3 \text{ s}^{-1}$. As for the time t , we are interested in the epoch when the temperature $T_\gamma \sim 10^9$ K. Using equation (10.122) for the density of relativistic particles, and following Gamow by assuming (not *quite* correctly) that this was all radiation with energy density $a_\gamma T_\gamma^4$ (a_γ is the standard blackbody radiation constant $7.5657 \times 10^{-16} \text{ J m}^{-3} \text{ K}^{-4}$), we find the time of interest t is

$$t = \frac{c}{T_\gamma^2} \left(\frac{3}{32\pi G a_\gamma} \right)^{1/2}, \quad (10.138)$$

which amounts to 230 s for $T_\gamma = 10^9$ K. This gives n very close to 10^{23} m^{-3} . Next, Gamow estimated a present-day average particle density of about 0.1 m^{-3} based on astronomical

estimates and observations, so that the Universe had expanded by a factor of 10^8 in the scale factor R . (Recall that the factor 10^8 reduces n by 10^{24} .) But an expansion of a factor of 10^8 in R means that the current T_γ should be near $10^9/10^8 = 10$ K! The millimeter-wave detectors that would have been required for this wavelength of observation were just at the leading edge of technology in the late 1940s, a by-product of the development of short-wavelength radar during the Second World War. Gamow could have, but did not, pursue this result aggressively, and the prediction was gradually forgotten. The CMB therefore could have been detected in the late 1940s, as part of the great postwar flowering of physics. But it wasn't.

This argument would be put slightly differently now, but in essence it is correct, a brilliantly simple piece of intuitive reasoning.

Exercise 10.8.2

Repeat the Gamow argument with modern cosmological numbers. Keep $T_\gamma = 10^9$ K and the σv value, but note that relativistic species present at the time include not only photons but three types of neutrinos and three types of antineutrinos, each neutrino-antineutrino combination contributing $(7/8)a_\gamma T_\gamma^4$ to the background energy density. We can neglect e^+ and e^- pairs in our relativistic fermion population. Why? The 7/8 factor arises because the neutrinos obey Fermi statistics. Recall from your statistical physics that the occupation number for a state with momentum p is

$$n_{\pm} = \frac{1}{\exp(pc/k_B T) \pm 1},$$

where the $+$ sign is for bosons (e.g., photons), and the $-$ sign, for fermions (e.g., neutrinos). The density of states (per unit volume) is the same for both types of particles: $4\pi p^2 dp/h^3$, where h is Planck's constant. The energy is the same for each type of particle, pc . Therefore, the difference in the total energy densities of a population of bosons or fermions comes down to the difference between two simple integrals:

$$\int_0^\infty \frac{x^3 dx}{e^x + 1} = \frac{7}{8} \int_0^\infty \frac{x^3 dx}{e^x - 1}$$

Can you prove this mathematically *without* doing either integral explicitly? (Hint: Consider the difference

$$\frac{1}{e^x - 1} - \frac{1}{e^x + 1},$$

and integrate x^3 times the residual.) Note that for each neutrino species the factor is actually 7/16, because only *one* spin helicity is present, as opposed to *two* spin states for photons. But then the neutrinos get back up to 7/8 with their antiparticle cousins . . . which the photons lack! You may take the current density in ordinary baryons to be 5% of the critical density $3H_0^2/8\pi G$.

Almost 20 years later, in 1965, the problem of determining the current T_γ attracted the attention of a team of physicists at Princeton University. They rediscovered for themselves the Gamow argument. The senior investigator, Robert Dicke, a highly regarded physicist both in theory and experiment, realized there was likely to be a measurable background radiation field that survived to the present day. Moreover, it could be rather easily detected by an instrument, the Dicke radiometer, that he himself had invented 20 years before in the postwar era! (A Dicke radiometer is a device that switches its view 100 times per second between the sky source and a carefully calibrated thermal heat bath of liquid helium. This imprints a 100 Hz Fourier component on the desired signal, and thereby reduces nuisance variability occurring on very different timescales.)

The team assembled: Dicke, the scientific leader; J. Peebles, a young theorist who would become the world's leading cosmologist; and P. Roll and D. Wilkinson, talented instrument builders who designed and built the contemporary Dicke radiometer. They were all set up to do the observation from the roof of their Princeton office building(!) when a phone call arrived from nearby Bell Telephone Laboratories. Two researchers, Arno Penzias and Robert Wilson, had found a puzzling extraneous signal in their own detector, a device designed to receive long-range signals relayed by what were some of the first commercial television satellites (the Telstar series). Naturally, these investigators were trying to chase down all possible sources of background confusion. The “effective radiation temperature” of the nuisance diffuse signal was about 3 K.¹³ Penzias and Wilson had no idea what to make of it, but they were advised by Bernard Burke, a radio astronomer colleague, to give the Dicke team at Princeton a call. The source of the excess radiation temperature was then quickly identified by the Princeton team.

The 1978 Nobel Prize in Physics went to Penzias and Wilson for their discovery of the cosmic microwave background radiation. In 2019, at the age of 84, Peebles was duly recognized with a Nobel Prize for his many contributions to modern cosmology.

10.8.3 The cosmic microwave background (CMB): Subsequent developments

The initial observations of the CMB were at wavelengths of 7 cm (Bell Labs) and 3 cm (Princeton). Needless to say, this does not constitute a very complete spectrum. The task of establishing the broader spectrum was fraught with difficulties, however, with many disputed and ultimately withdrawn claims of large deviations from a blackbody. The formal Wien peak of the CMB is close to 1 mm, and to access shorter wavelengths at the desired level of accuracy required satellite-based cryogenic receivers. In 1992, matters were finally laid to rest when the COBE satellite returned its dramatically undramatic finding that the CMB is, very nearly, but not *exactly*, a blackbody. There are small fluctuations in the temperature, the largest of which amount to a few parts in 10^4 . This value was smaller than first expected but ultimately understood to be of the order needed to account for the nonlinear structure in the Universe that we see today, provided there is a healthy component of *dark matter* present that does not react with the radiation. Having

¹³A single observation of the radiation intensity is enough to determine the entire spectrum if the source is known with certainty to be a blackbody. Hence, the data point of Penzias and Wilson would have come from a 3 K blackbody, if that indeed was the source.

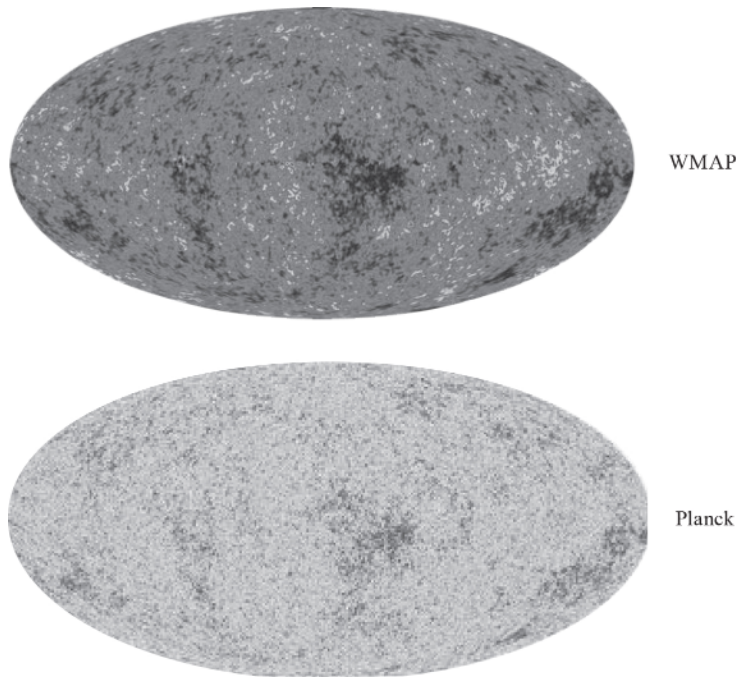


Figure 10.11 WMAP (top) and Planck (bottom) temperature fluctuations, with different color scales. Think of these as the Universe’s baby pictures. Note the much higher angular resolution compared with those obtained by COBE; Planck is another factor of 3 better than WMAP. The angular information encoded on these small scales is valuable as a means of tightly constraining cosmological parameters.

been invoked decades earlier to account for the dynamic structure of galaxies and clusters of galaxies, dark matter’s mysterious presence was further strengthened by the COBE results.

Following COBE, the next important CMB probe was the Wilkinson Microwave Anisotropy Probe, or WMAP. This is named for David Wilkinson, the same Princeton researcher who had been instrumental in the earliest CMB studies. With the launch of WMAP in 2003, cosmology truly became a precision science. WMAP revealed the structure of the temperature fluctuations in such exquisitely fine angular resolution on the sky, it became possible to determine all the key physical parameters of the Universe: H_0 , Ω_M , Ω_V , t_0 , and many others, to an excellent level of accuracy. WMAP was followed by the Planck satellite, launched in 2009, which provided a further shrinking of the error bars, higher angular resolution coverage of the CMB on the sky, broader frequency coverage (very important for subtracting the effects of the Milky Way Galaxy), and better constraints on null results (absence of CMB polarization, for example). But there were no qualitatively new physical findings. The story emerging from the Planck CMB probe remained essentially the one revealed to us by WMAP (figure [10.11]), but on a yet-stronger footing.

After the breakthrough findings, the supernova studies continued to evolve in both quantity and quality. Here is where the story becomes interesting, because for many years there seemed to be significant statistical discrepancy between the Hubble constant inferred from

the supernova work¹⁴ ($72 \text{ km s}^{-1} \text{ Mpc}^{-1}$), and the CMB probes¹⁵ ($67 \text{ km s}^{-1} \text{ Mpc}^{-1}$). The claimed accuracies were such that this amounted to a five standard deviation (5σ) discrepancy. It is to be noted that the CMB value is inferred from early times in the history of the Universe, whereas the supernova measurements are much more recent. Taken at face value, this meant that either at least one of these values was incorrect or the Λ parameter was not, in fact, a constant after all! The latter would require new physics: in standard general relativity, Λ must be rigorously constant.

Very recent work, however, suggests the tension may be heading toward a resolution, though arguments continue.¹⁶ The remarkable sensitivity and wavelength coverage of the James Webb Space Telescope allows stellar populations to be used as luminosity standards with tighter error control. The value found by Freedman et al. (2025) of ($70.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$) appears to be compatible with both early and recent time H_0 measurements, while remaining consistent with standard, constant Λ cosmology as well. Additional data from Webb will increase the precision yet further.

Think of the evolution of the maps and globes of Earth, from ancient to modern times. Gross inaccuracies in basic geometry gradually evolved to near-perfect precision, but only over a period of thousands of years. Contrast this with the observation that, within the lifetimes of currently active researchers, serious models of the Universe went from fundamental misconceptions to a few percent accuracy in structural parameters. By any measure, this is one of the great scientific achievements of our time. As with any great scientific advance, its full meaning and ultimate implications will take many years to elucidate.

10.9 Thermal history of the Universe

10.9.1 Preliminaries

The recorded history of civilization began when Sumerian merchants inscribed their grain inventories on reusable clay tablets. (The concept of a “grain inventory” would not have been possible without the invention of large-scale agriculture, which would not have been possible without the invention of astronomy. But we digress.) Etched in the matrix of clay, these markings became frozen in time because the clay, once dried, remained unaltered and intact. It turns out that the Universe has its own gigantic clay tablet, which allowed inscriptions first to be imprinted, then to be preserved, more-or-less unaltered for 13.8 billion years. We know these inscriptions as temperature fluctuations, and the clay tablet is the cosmic microwave background.

The temperature fluctuations trace the initial density fluctuations, which we have seen are coupled to the radiation by electron-photon scattering, also known as *Thomson scattering*. While this scattering remained vigorous, the CMB dutifully recorded and rerecorded the ever-evolving changes in the density. Then, matter and radiation decoupled from one another. The pattern imprinted on the CMB abruptly stopped being erased and recorded anew. The CMB instead retained only the very last pattern that happened to be imprinted upon it, at the time of the “last scattering.” It is *this* pattern that we receive, redshifted by the enormous subsequent expansion of the Universe, in our detectors today. Think of this either

¹⁴Riess et al. 2022 *ApJ*, 934, L7.

¹⁵Planck Collaboration et al. 2020, *Astron. Astrophys.*, 641, A6.

¹⁶Freedman, W.; et al. 2025, arXiv 2408.6153v3 [astro-ph.CO].

as an ancient inscription passed on through the eons, or more congenially as the Universe's baby picture.

Exercise 10.9.1

Let σ_T be the Thomson cross section for a photon to collide with an electron, a constant value equal to $6.7 \times 10^{-29} \text{ m}^2$. When an electron has moved relative to the photon gas a distance l such that the swept-out volume $l\sigma_T$ captures a single photon, l is one *mean free path* (mfp): the average distance between scatterings.

Justify this definition, and show that the scattering rate per electron is $n_\gamma c \sigma_T = c/l$. (What is n_γ here?) In a radiation-dominated Universe, show that the ratio of the photon scattering mfp to the horizon size grows as $\sqrt{t}$, and that the mfp relative to the scale factor R grows as t (as usual, time).

Exactly how to decipher the ancient CMB inscriptions and turn them into a model of the Universe is a very complicated business, one that we will be able to treat only very superficially here. We shall go about this in two stages. In the first, we will describe the classical theory of the thermal history of the Universe: how matter and radiation behaved in one another's presence from temperatures of 10^{12} K through 3000 K , the time when hydrogen ions recombined with electrons to form neutral hydrogen atoms. In the second stage, reserved for chapter 11, we will discuss a more modern theory of the very early Universe. This puts a premium on the notion of *inflation*, an early period in the history of the Universe in which it seems to have undergone a very rapid (exponential) growth phase. First put forth in 1981, this idea, generally credited to Alan Guth (who coined the term “inflation”), has arguably been the single most important theoretical advance in modern cosmology in the last half century.¹⁷ There are good theoretical reasons for invoking the process of inflation, even if the detailed mechanism is not well understood (hardly an unusual state of affairs in science), and there is by now compelling observational support for it. One of the best arguments for inflation as a concept is that we have, in fact, entered another, albeit far milder, inflationary stage of the Universe's history.

In what follows, the reader should bear in mind the characteristic physical quantities on what is known as the “Planck scale.” These are the quantities set by the three fundamental dimensional constants of physics: Newton's G , Planck's h , and the speed of light c . There are unique dimensional combinations for forming a mass m_P , a length l_P , and a time t_P from these constants:

$$m_P = \left(\frac{hc}{G} \right)^{1/2} = 5.456 \times 10^{-8} \text{ kg}, \quad (10.139)$$

$$l_P = \left(\frac{hG}{c^3} \right)^{1/2} = 4.051 \times 10^{-35} \text{ m}, \quad (10.140)$$

$$t_P = l_P/c = \left(\frac{hG}{c^5} \right)^{1/2} = 1.351 \times 10^{-43} \text{ s}. \quad (10.141)$$

¹⁷Guth, A. H. 1981, *Phys. Rev. D*, 23, 2, 347.

With these three quantities, we can form any other dimensional quantity we like. These are, in some sense, the limits of our knowledge, for it is on these scales that we may expect quantum gravity effects to be important, a theory of which we remain quite ignorant. We cannot expect to have anything like a classical picture of the early Universe for time $t < t_p$ or for horizon scales $c/H < l_p$. The Planck mass may not, at first glance, appear remarkable, but remember, the natural comparison is with elementary particle masses. An electron, for example, is close to 5×10^{-4} GeV in rest energy; the Planck mass energy is $m_p c^2 = 3.06 \times 10^{19}$ GeV. By these Planckian standards, we will always be working with very low mass particles, very large distances, and very long times. However, not every Planck-scale quantity pushes the boundaries of known physics. The Planck velocity is, after all, just the speed of light. That, we can accommodate!

10.9.2 Classical cosmology: Helium nucleosynthesis

We begin the story when the temperature of the Universe is just below 10^{12} K. This is very early on, perhaps 10^{-4} s after the Big Bang. At this stage, the Universe consists of a relativistic cocktail of photons, neutrinos and their antiparticles, muons and their antiparticles, and electrons and their positron antiparticles. Even the neutrinos are in complete collisional thermal equilibrium, freely created and destroyed.¹⁸ There is also a population of protons and neutrons, which are energetically unimportant at these temperatures. But keep track of them in the inventory: they are going to make the Universe we know as home.

Once the temperature slips below 10^{12} K, the muons and antimuons can annihilate, but there isn't enough energy for muon production from the other relativistic populations present. The energy from the photons and e^+e^- pairs produced in the annihilation is shared among all the other populations, including, at these ultrahigh densities, the normally elusive neutrinos.

The Universe continues to expand; the temperature falls. To understand the next important phase, recall that neutrons and protons do not have quite the same mass. More precisely, with m_p the proton mass and m_n the neutron mass,

$$\Delta mc^2 \equiv (m_n - m_p)c^2 = 1.293 \text{ MeV} = 2.072 \times 10^{-13} \text{ J}. \quad (10.142)$$

This is very small compared with either $m_n c^2 = 939.265 \text{ MeV}$ or $m_p c^2 = 938.272 \text{ MeV}$. Except when we are directly concerned with the mass difference itself or doing very accurate calculations, there is normally no need to distinguish between m_n and m_p . The ratio of the probability of finding the system in i (denoted by P_i) to the probability of finding it in j (P_j) is given by the Boltzmann equation:

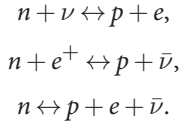
$$\frac{P_i}{P_j} = \frac{g_i}{g_j} \exp \left[- \left(\frac{E_i - E_j}{k_B T} \right) \right]. \quad (10.143)$$

Here, the g 's are the *statistical weights*, the number of distinct states available with energy E_i or E_j . Below 10^{11} K, the exponential Boltzmann factor $\exp(-\Delta mc^2/k_B T)$ starts to differ noticeably from unity, and as we approach $T = 10^{10}$, significantly so. Since neutrons are

¹⁸By using the word “freely,” we ignore the particle rest masses.

more massive than protons, the former are more scarce at cooler temperatures. The critical temperature, which corresponds to $k_B T = \Delta mc^2$, is 1.5×10^{10} K. (This is not coincidentally very comparable to $m_e c^2 / k_B = 5.9 \times 10^9$ K.)

The reactions that determine the neutron n to proton p balance are ($\bar{\nu}$ denotes an antineutrino):



The reaction rates for the two-body processes are about 0.1 s^{-1} per nucleon at $T = 10^{10}$ K, rapidly rising above this temperature and rapidly declining as the temperature falls. Below 10^{10} K, the reactions cannot keep up with the expansion rate of the Universe; they simply do not take place fast enough. Once this occurs, whatever the ratio of n/p happens to be, it remains temporarily “frozen” with time, since the Universe is too cold to keep the two-body reactions cooking! At $T = 10^{10}$ K, the Universe is about 1 s old. The n/p ratio at this temperature is $\exp(-\Delta mc^2 / k_B T) \simeq 0.2$, and there it remains for a relatively extended duration.

Exercise 10.9.2

Show that the early Universe “temperature clock” is given conveniently by

$$t \simeq 1/T_{10}^2,$$

where t is the time in seconds and T_{10} the temperature in 10^{10} K. Assume a Universe consisting of photons, electrons and positrons, and three neutrino and antineutrino families.

This value of 0.2 is interesting, because it is neither close to unity nor tiny (a coincidence related to the observed 25% helium mass fraction). Without further production of neutrons, they will decay over a timescale of minutes by the third reaction channel. If all the neutrons had decayed into protons, electrons, and antineutrinos, there would have been no subsequent cosmological nucleosynthesis. But just as the neutrons start to decay as we approach 10^9 K, they are salvaged, safely packed into stable ^4He nuclei. Helium nuclei are environmentally sound containment vessels for otherwise radioactive neutrons. In ^4He nuclei, neutron decay is absolutely forbidden, in essence because the available statistical phase space for decay within the realm of the nuclear potential is degenerate: there are no available states for the neutrons to decay into. They are already occupied by protons.

The neutrons are now quickly scarfed up into helium nuclei. This offers a firm prediction for the observed mass fraction of the Universe, since later stellar nucleosynthesis does not change the helium number significantly. (By sharp contrast, almost all the heavier element abundances, collectively not much more than 1% of the total, are dominated by stellar nucleosynthesis. If helium were made this way, its mass fraction would also be at the percent

level.) From what we have just seen, we may estimate the mass fraction in helium to be

$$\frac{4m_p \times (n/2)}{m_p(n+p)} = \frac{2}{1+p/n} \simeq 0.33. \quad (10.144)$$

Much more detailed, time-dependent calculations (pioneered in a classic paper by Peebles¹⁹) give a number close to 0.27 due to neutron decay (a slight refinement of the 25% figure mentioned above), but the essential physics is captured by our simple estimate. There is not very much wiggle room here. We cannot use the measured value of the helium abundance to determine retrospectively what sort of Universe (open, closed, critical) we live in. The curse and strength of the calculation is that the results are essentially the same for *any* FRW model. The Big Bang theory robustly predicts something close to 27% of the mass of the Universe is in the form of helium, and that is that. Happily, this turns out to be very close to what observations reveal.

10.9.3 Neutrino and photon temperatures

As the temperature slides from 5×10^9 K to below 10^9 K, the electron-positron pairs annihilate into photons:

$$e^+ + e^- \rightarrow 2\gamma.$$

The falling temperature prevents the photons from maintaining an equilibrium population of these e^+e^- pairs. There is then a net conversion of electrons and positrons into photons. However, it is not possible to just add these photons and keep the temperature unchanged. In thermal equilibrium, the number of photons at a given temperature is *fixed*. If you add more via e^+e^- annihilation, the extra photons force a new thermal equilibrium, one at a higher temperature that is compatible with the increased photon number. The photons are, in effect, heated. By contrast, the relativistic neutrinos, with their tiny cross sections, just march along with the expansion of the Universe, oblivious to everything. (By contrast, when it was the muons that annihilated, the Universe was so dense that even the neutrinos took part in the thermal collisions. Not this time!) This means that present-day cosmic background neutrinos should be somewhat cooler than the CMB photons. The question is, by how much?

This turns out to be a relatively simple problem, because if we fix our attention on a comoving volume of the Universe, the entropy in this volume is conserved by the conversion of electron-positron pairs into photons. For photons or relativistic electrons/positrons, the entropy per unit volume is a function only of the temperature. It is most easily calculated from a standard thermodynamic identity,

$$E + PV - TS + \underbrace{\mu N}_0 = 0,$$

for a gas with zero chemical potential μ , which is true of a relativistic gas that freely creates and annihilates its own particles. The entropy per unit volume is then

$$\frac{S}{V} \equiv s = \frac{E}{VT} + \frac{P}{T}. \quad (10.145)$$

¹⁹Peebles, P.J.E. 1966, *ApJ*, 146, 542.

With ρc^2 denoting the energy per unit volume and $P = \rho c^2/3$ for a relativistic gas,

$$s = \frac{4\rho c^2}{3T}. \quad (10.146)$$

Since any equilibrium relativistic energy density is always proportional to T^4 , the entropy per unit volume is proportional to T^3 , and the entropy within a comoving volume R^3 is proportional to $(TR)^3$. It is conserved with the expansion, in essence because the entropy and particle number are proportional to each other, and overall particle number is conserved.

Before the e^+e^- annihilation, the entropy within a dimensionless volume R^3 in photons, electrons, and positrons is

$$sR^3 = \frac{4a_\gamma(TR)^3}{3} \left(1 + 2 \times \frac{7}{8}\right) = \frac{11a_\gamma(TR)_{\text{before}}^3}{3}. \quad (10.147)$$

After the annihilation, and after the reestablishment of thermal equilibrium, it is all photons:

$$sR^3 = \frac{4a_\gamma(TR)_{\text{after}}^3}{3}. \quad (10.148)$$

Since the number of photons produced is equal to the number of relativistic electrons and positrons that annihilated, this process conserves entropy, sR^3 . The process is reversible: one could recompress the adiabatic expansion backward to produce anew the pairs. Since R itself is continuous,

$$\frac{4T_{\text{(after)}}^3}{3} = \frac{11T_{\text{(before)}}^3}{3}. \quad (10.149)$$

The photons are, in effect, heated.

Now for an interesting point of physics. The earlier muon-antimuon annihilation heated *all* relativistic populations because the reaction rates were high when the Universe was hotter and denser. But the e^+e^- annihilation we are now considering ignores the neutrinos, because the density now has fallen to a level at which the neutrinos pass through everything. *Their* temperature does not increase during the e^+e^- annihilation, and this process will therefore produce different cosmic background photon and neutrino temperatures. The ratio of neutrino temperature T_ν to photon temperature T_γ now follows from

$$T_\nu(\text{after}) = T_\nu(\text{before}) = T_\gamma(\text{before}) = \left(\frac{4}{11}\right)^{1/3} T_\gamma(\text{after}) = 0.7138 T_\gamma(\text{after}). \quad (10.150)$$

This result would correspond to a current value of $T_\nu = 1.95$ K—if the neutrinos remained a $\mu = 0$ relativistic population for all time. We now know they do not—at least not quite. At one time, people wondered whether there was any possible way to measure this temperature difference. But a temperature of 1.95 K corresponds to an energy kT of 1.7×10^{-4} eV, whereas the average mass per neutrino species (as best we can currently measure) is about 0.1 eV, with a corresponding temperature of 1160 K. Neutrinos thus became “cold” at a redshift of $z \sim 400$. As we shall see, this is well after hydrogen recombined but almost certainly before galaxies formed. These neutrinos are a part, albeit a very small part, of the dark

matter in galaxies. Even if it *had* turned out that neutrinos had zero rest mass, it would have been impossible with present technology to measure a 2 K neutrino background! Nevertheless, the physics of this problem is enlightening, and the formal difference in neutrino and photon temperatures, surprising.

10.9.4 Ionization of hydrogen

Between the time of helium synthesis, a few minutes into the Universe's life, and a few hundred thousand years later, almost nothing happens. Radiation remains the dominant source of energy and pressure, and the Universe simply expands with R scaling as $t^{1/2}$. The matter and the radiation remain tightly coupled, so that density fluctuations in the matter correspond also to energy fluctuations in the radiation. But at some point, the Universe becomes sufficiently cool that hydrogen recombines. The matter ceases to be an ionized plasma and becomes a neutral gas. At what temperature or, equivalently, at what redshift does this occur? The answer to this question follows from the Saha equation, which in turn follows from an interesting generalization of the Boltzmann equation. The Saha equation determines the ionization fraction of an atomic element in thermal equilibrium (an important assumption!) as a function of the temperature T .

To give a simple derivation, recall the Boltzmann equation, (10.143): the probability P_i for an atom to be in state i relative to the probability P_j to be in state j is

$$\frac{P_i}{P_j} = \frac{g_i}{g_j} \exp\left(-\frac{E_i - E_j}{k_B T}\right), \quad (10.151)$$

where E_i, E_j are the energies of the respective states, and g_i, g_j the statistical weights. For simplicity, we consider a gas of pure hydrogen, and interpret this equation as follows. State j is the ground state of hydrogen with one electron, the “1s” state. We set $j = 0$ for this neutral state. State i is then the *ionized* state with a bare proton and a free electron with kinetic energy between E and $E + dE$. We set $i = 1$. Whether bound or free, the electron always has two spin states available to it, so this factor of 2 appears in both numerator and denominator of the statistical weight factor, canceling in the process. Thus, ignoring spin, g_0 is unity, and g_1 is the number of states available to the electron with energy in the range dE , as per above. This is the interesting part of the calculation. In terms of the electron momentum p (a scalar magnitude here), the number of states per electron within a shell of thickness dp in momentum space is

$$\frac{4\pi p^2}{n_e h^3} dp,$$

where the inverse number density factor $1/n_e$ represents the volume per electron, and h is Planck's constant. (This is derived in any standard text on statistical mechanics. The result follows from the stratagem of putting the electrons in a large cube of volume V , counting the eigenstates for each electron, and then noting that the artificial box appears in the calculation only in the form of a volume per electron, which is just $1/n_e$.) When we draw all these threads together and sum over all possible free electron states, equation (10.151) becomes

$$\frac{n_e n_p}{n_0} = \exp\left(-\frac{\Phi}{k_B T}\right) \frac{4\pi}{h^3} \int_0^\infty p^2 e^{-p^2/2m_e k_B T} dp, \quad (10.152)$$

where Φ is the ionization potential of hydrogen, m_e the electron mass, n_p the proton density, and n_0 the neutral hydrogen density. Note that we have replaced the ratio of relative probabilities of the ionized hydrogen state (or p) to the neutral hydrogen state (subscript 0) with their number densities n_p and n_0 . Now

$$\int_0^\infty p^2 e^{-p^2/2m_e k_B T} dp = (2m_e k_B T)^{3/2} \int_0^\infty x^2 e^{-x^2} dx = (2m_e k_B T)^{3/2} \times \frac{\pi^{1/2}}{4}. \quad (10.153)$$

The Saha equation is then

$$\frac{n_e n_p}{n_0} = \left(\frac{2\pi m_e k_B T}{h^2} \right)^{3/2} \exp \left(-\frac{\Phi}{k_B T} \right). \quad (10.154)$$

The final step is to note that for pure hydrogen, $n_e = n_p$ and thus $n_e + n_0$ is the total hydrogen density n_H . With the ionization fraction $x = n_e/n_H$, our final equation becomes

$$\frac{x^2}{1-x} = \frac{1}{n_H} \left(\frac{2\pi m_e k_B T}{h^2} \right)^{3/2} \exp \left(-\frac{\Phi}{k_B T} \right) = \left(\frac{2.415 \times 10^{21} T^{3/2}}{n_H} \right) \exp \left(-\frac{1.578 \times 10^5}{T} \right), \quad (10.155)$$

in SI units.

Let us take a nominal value of $x = 0.5$ for the onset of ionization. The ambient temperature is given by $T = T_{\gamma 0}(1+z)$, where $T_{\gamma 0}$ is the current radiation temperature of 2.732 K. The current best Planck CMB data suggest an average baryon mass density of about $4.6 \times 10^{-28} \text{ kg m}^{-3}$. In a pure hydrogen gas that would correspond to $0.275 \text{ nuclei m}^{-3}$. With these numbers, we find that the onset of ionization is attained at a redshift of about 1400, and a temperature of 3800 K. This is naturally close to the formal Thomson scattering decoupling redshift, for which detailed calculations give $z \simeq 1100$. (The helium contribution changes these numbers only modestly, because the equation is largely controlled by the exponential Boltzmann factor. Note that the ionization transition occurs well below the formal Boltzmann ionization temperature of $1.58 \times 10^5 \text{ K}$.) At redshifts less than 1100, the Universe therefore becomes transparent to photons; the total energy density is already dominated by matter. Interestingly, however, the intergalactic medium seems to have been completely *reionized* shortly after galaxies were able to form at $z \sim 10$, presumably by the radiation produced by the accretion process that gave rise to these galaxies, as well as by the active galactic nuclei they contain. Understanding how the Universe became reionized is a lively area of current astrophysical research.